MATERIALS SCIENCE

A catalyst family of high-entropy alloy atomic layers with square atomic arrangements comprising iron- and platinum-group metals

Cheng-Yu Wu[1], Yueh-Chun Hsiao[1], Yi Chen[1], Kun-Han Lin[1], Tsung-Ju Lee[2], Chong-Chi Chi[3], Jui-Tai Lin[1], Liang-Ching Hsu[4], Hsin-Jung Tsai[2], Jia-Qi Gao[1], Chun-Wei Chang[1], I-Ting Kao[1], Chia-Ying Wu[1], Ying-Rui Lu[4], Chih-Wen Pao[4], Sung-Fu Hung[2,5], Ming-Yen Lu[3], Shan Zhou[6], Tung-Han Yang[1,7]*

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We report a catalyst family of high-entropy alloy (HEA) atomic layers having three elements from iron-group metals (IGMs) and two elements from platinum-group metals (PGMs). Ten distinct quinary compositions of IGM-PGM-HEA with precisely controlled square atomic arrangements are used to explore their impact on hydrogen evolution reaction (HER) and hydrogen oxidation reaction (HOR). The PtRuFeCoNi atomic layers perform enhanced catalytic activity and durability toward HER and HOR when benchmarked against the other IGM-PGM-HEA and commercial Pt/C catalysts. Operando synchrotron x-ray absorption spectroscopy and density functional theory simulations confirm the cocktail effect arising from the multielement composition. This effect optimizes hydrogen-adsorption free energy and contributes to the remarkable catalytic activity observed in PtRuFeCoNi. In situ electron microscopy captures the phase transformation of metastable PtRuFeCoNi during the annealing process. They transform from random atomic mixing (25°C), to ordered $L1_0$ (300°C) and $L1_2$ (400°C) intermetallic, and finally phase-separated states (500°C).

INTRODUCTION

Since the first report on solid-solution high-entropy alloys (HEAs) was published in 2004, these materials have been regarded as a paradigm in a wide range of applications (*1*, *2*). Recently, dozens of single-phase HEA nanocrystals or nanostructures and their corresponding intermetallic and oxide derivatives have been developed for various catalytic applications, demonstrating superior activity, selectivity, and durability (*3*–*21*). A series of nanoscale HEA catalysts mainly composed of platinum-group metals (PGMs: Pt, Pd, Rh, Ru, Ir, and Os) have been considered the prime choice (*8*, *17*, *20*). It is because each PGM can promote distinctive reactions, and, thus, PGM-HEA would create an ideal surface for complex or multistep catalysis. For instance, the PtPdRhRuIrOs HEA nanocrystals are one of the best catalysts for ethanol oxidation reaction with 12-electron/12-proton transfer processes (*8*). However, since PGMs are scarce, minimizing the demand for PGMs is extremely important. To address this issue, alloying PGMs with earth-abundant iron-group metals (IGMs: Fe, Co, and Ni), which are the neighboring elements with PGMs on the periodic table, into single-phase HEA nanocrystals (referred to as IGM-PGM-HEA nanocrystals; table S1) has attracted much attention (*5*, *7*, *10*, *11*, *14*, *18*). The IGM-PGM-HEA nanocrystals as catalysts not only can reduce the usage of PGMs but also generate the synergistic effect (or called the "cocktail effect") among constituent elements to boost catalytic performance. Note that the cocktail effect is one of the four core effects in the field of high-entropy materials, indicating that inter-element interactions between the constituent elements give rise to remarkable and/or unusual performances (*1*, *2*). The local lattice distortion and electronic interactions between these 3d, 4d, and 5d transition metals within the IGM-PGM-HEA lattices may achieve competitive preferential adsorption sites for the reactants, intermediates, and/or products, which endow enhanced catalytic performance (*22*, *23*).

Despite the recent surge in IGM-PGM-HEA–based research, several great challenges remain:

1.It is hard for the current synthetic methods to control the surface atomic structures of IGM-PGM-HEA nanocrystals, and, thus, facet-controlled IGM-PGM-HEA nanocrystals are lacking, which is expected to affect catalytic performance. In particular, IGMs and PGMs are crystallized in one of the three different crystal lattices, including body-centered cubic, face-centered cubic (FCC), and hexagonal close-packed at the thermodynamic states (table S2). Therefore, it is a challenge to confine five or more elements of them on the nanocrystal surface with a specific arrangement.

2.Obtaining an optimized composition of IGM-PGM-HEA nanocrystals for catalysis remains an issue. This is due to the lack of a comprehensive understanding of the catalytic performances of IGM-PGM-HEA nanocrystals with various combinations of different metals at the same surface atomic structures.

3.A knowledge gap exists in how a synergistic effect among IGM and PGM elements improves catalytic performance. An in-depth understanding of the synergistic interactions in a high-entropy multielemental phase as well as their actual active sites during catalysis is needed by operando characterization tools.

4.There is an urgency to identify whether IGM-PGM-HEA nanocrystals are thermodynamically stable or kinetically favored phase

[1]Department of Chemical Engineering, National Tsing Hua University, Hsinchu 30013, Taiwan. [2]Department of Applied Chemistry, National Yang Ming Chiao Tung University, Hsinchu 30010, Taiwan. [3]Department of Materials Science and Engineering, National Tsing Hua University, Hsinchu 30013, Taiwan. [4]National Synchrotron Radiation Research Center, Hsinchu 30076, Taiwan. [5]Center for Emergent Functional Matter Science, National Yang Ming Chiao Tung University, Hsinchu 300093, Taiwan. [6]Department of Nanoscience and Biomedical Engineering, South Dakota School of Mines and Technology, Rapid City, SD 57701, USA. [7]High Entropy Materials Center, National Tsing Hua University, Hsinchu 30013, Taiwan.
*Corresponding author. Email: tunghanyang@mx.nthu.edu.tw

(metastable phase), which will provide a fundamental perspective on the stability of a high-entropy single phase (*24*–*26*).

To conquer these challenges, herein, we synthesize a catalyst family of nanocrystals exposing well-defined IGM-PGM-HEA facets using wet-chemical synthesis. As a proof of concept, we combine three elements from IGMs (i.e., Fe, Co, and Ni) with two elements from PGMs (i.e., Pt, Pd, Rh, Ru, and Ir) to prepare 10 different quinary IGMPGM-HEA subnanometer-thick atomic layers on the Pd seeds with a well-defined shape, such as Pd@PtRuFeCoNi core-shell nanocubes enclosed by the square atomic arrangements of {100} facets. We systematically examine the catalytic performances of IGM-PGM-HEA atomic layers toward hydrogen evolution reaction (HER) and hydrogen oxidation reaction (HOR). We also elaborate on the corresponding atomic-scale synergistic interactions, active sites, and catalytic mechanisms during catalysis through operando synchrotron x-ray absorption spectroscopy (XAS) and density functional theory (DFT) simulations. Furthermore, we use in situ high-angle annular dark-field scanning transmission electron microscopy (HAADF-STEM) to confirm the metastable phase of IGM-PGM-HEA atomic layers and reveal their equilibration pathways.

RESULTS

Synthesis and characterization of IGM-PGM-HEA atomic layers

In recent years, many alloy catalysts covered by the {100} facets as the predominant surface atomic structure have shown outstanding electrocatalytic activity and durability for HER (*27*–*31*). On the basis of these findings, as shown in Fig. 1, here, we choose to focus on the epitaxial growth of quinary IGM-PGM-HEA atomic layers on the FCC Pd cubic seeds (fig. S1) to form Pd@IGM-PGM-HEA core-shell nanocubes enclosed by the {100} facets. Figure 1A shows a schematic of the periodic table, highlighting the chemical combination of all 10 IGM-PGM-HEA atomic layers prepared. In our study, the Pd@PtRuFeCoNi core-shell nanocubes are first demonstrated (see Materials and Methods for details), as shown in Fig. 1B. Briefly, the PtRuFeCoNi shells can be deposited on the Pd nanocubes by adding the precursor solution, including H_2PtCl_6, $RuCl_3{\cdot}xH_2O$, $Fe(C_5H_7O_2)_3$, $Co(C_5H_7O_2)_2$, and $Ni(C_5H_7O_2)_2$ at a slow rate of 0.8 ml hour^{-1} to the reaction solution of oleylamine (Oam) and 1-octadecene (ODE) containing the preformed Pd cubic seeds at a high enough temperature of 200°C. Such dropwise introduction of precursor solution can allow us to control the number of atoms generated from the five precursors at about the same value during the steady state for the number of each precursor (*20*, *32*), enabling the epitaxial growth of homogeneous PtRuFeCoNi as well as the elimination of self-nucleation process and thus island growth (fig. S2). Note that the layer-by-layer growth mode of PtRuFeCoNi atomic layers shifts to island growth when using Pd octahedral seeds enclosed by {111} facets, with the same injection rate of 0.8 or slightly lower, such as 0.6 ml hour^{-1}. This change is attributed to the predominant region of the deposited Pt, Ru, Fe, Co, and Ni atoms on the {100} facets, rather than the {111} facets, due to the elevated surface energy associated with the {100} facets characterized by square atomic arrangements compared to the {111} facets with hexagonal atomic arrangements (*33*). Further reduction of the injection rate to 0.4 ml hour^{-1} and an extension of the growth time to 36 hours result in a transformation in the products of seed-mediated growth, producing a more uniform surface (fig. S6). This observation suggests that when the rate of diffusion is sufficiently high compared to the rate of deposition, a more complete and conformal shell forms on the Pd octahedral seeds enclosed by the {111} facets while maintaining the octahedral morphology. These results confirm the generality of epitaxial growth developed in this study. Similar protocols are carried out for the preparations of the other nine types of Pd@IGM-PGM-HEA core-shell nanocubes with different compositions except for changing the types of the metal precursors.

Figure 1 (C to J) shows the structural and compositional characterizations of PtRuFeCoNi atomic layers on the 16.4-nm Pd cubic seeds (fig. S1). TEM image reveals good retention of cubic shape after the deposition (Fig. 1C). The edge length of Pd@PtRuFeCoNi core-shell nanocubes is increased to 18.4 nm (Fig. 1I), indicating around 1.0-nm thickness of PtRuFeCoNi shell layers. HAADF-STEM image taken from an individual nanocube reveals the periodic lattice extending across the whole nanocube covered by {100} facets, suggesting the epitaxial growth mode (Fig. 1D). An obvious contrast is observed on the outer surface consisting of a few atomic layers, which can be attributed to the difference in atomic number between these five elements. Smaller atoms with less contrast can be attributed to the lighter elements of IGMs (*9*, *21*, *24*). From the atomic-resolution HAADF-STEM image, together with the corresponding fast Fourier transform (FFT) pattern and intensity profile (Fig. 1, E and F, and fig. S7), the nanocube only has one set of an FCC-diffraction single-crystal pattern. The lattice spacing of (200) plane is around 0.20 nm. X-ray diffraction (XRD) analysis also confirms the diffraction peaks of the FCC phase (Fig. 1H). When compared to that of Pd nanocubes, the diffraction peaks of the Pd@PtRuFeCoNi core-shell nanocubes shift toward lower 2θ values. In addition, the full width at half maximums (FWHM) of the Pd@PtRuFeCoNi core-shell nanocubes (FWHM = 1.01) is found to be slightly larger than that of the Pd nanocubes (FWHM = 0.81), contrary to the expected trend based on the Scherrer equation. This slight deviation can be ascribed to various factors extending beyond crystallite size. One possible explanation for this deviation is that the Scherrer equation assumes uniform crystallite size and may not fully account for structural complexities, such as the core-shell nature and local lattice distortion with inhomogeneous strain introduced by the formation of the HEA shells consisting of disparate atomic sizes of Pt (2.774 Å), Ru (2.718 Å), Fe (2.475 Å), Co (2.505 Å), and Ni (2.530 Å) atoms (*2*). These crystal lattice imperfections may play a crucial role in contributing to XRD peak broadening. In addition, the average number of PtRuFeCoNi atomic layers on the Pd seeds is estimated as about five atomic layers from the atomic ratio of core and shell (table S3), in good agreement with the observation of ~1.0-nm PtRuFeCoNi thickness in the TEM image (Fig. 1, C and I).

Furthermore, the energy-dispersive x-ray spectrometry (EDS) elemental mappings (Fig. 1G) and their corresponding line scan profiles (fig. S8) verify the spatial core-shell distributions of Pd@PtRuFeCoNi nanocubes, with the core composed of Pd and solid-solution shell composed of five elements. The atomic percentages of Pt, Ru, Fe, Co, and Ni elements are 20, 22, 21, 19, and 18%, respectively, confirmed by inductively coupled plasma optical emission spectrometry (ICP-OES) analysis (Fig. 1J). Nevertheless, the EDS signals from IGMs including Fe, Co, and Ni exhibit relatively lower intensity when compared to those of PGMs comprising Pt and Ru in the HEA atomic layers, which shows the discrepancies in composition between the ICP-OES measurement (an overall measurement)

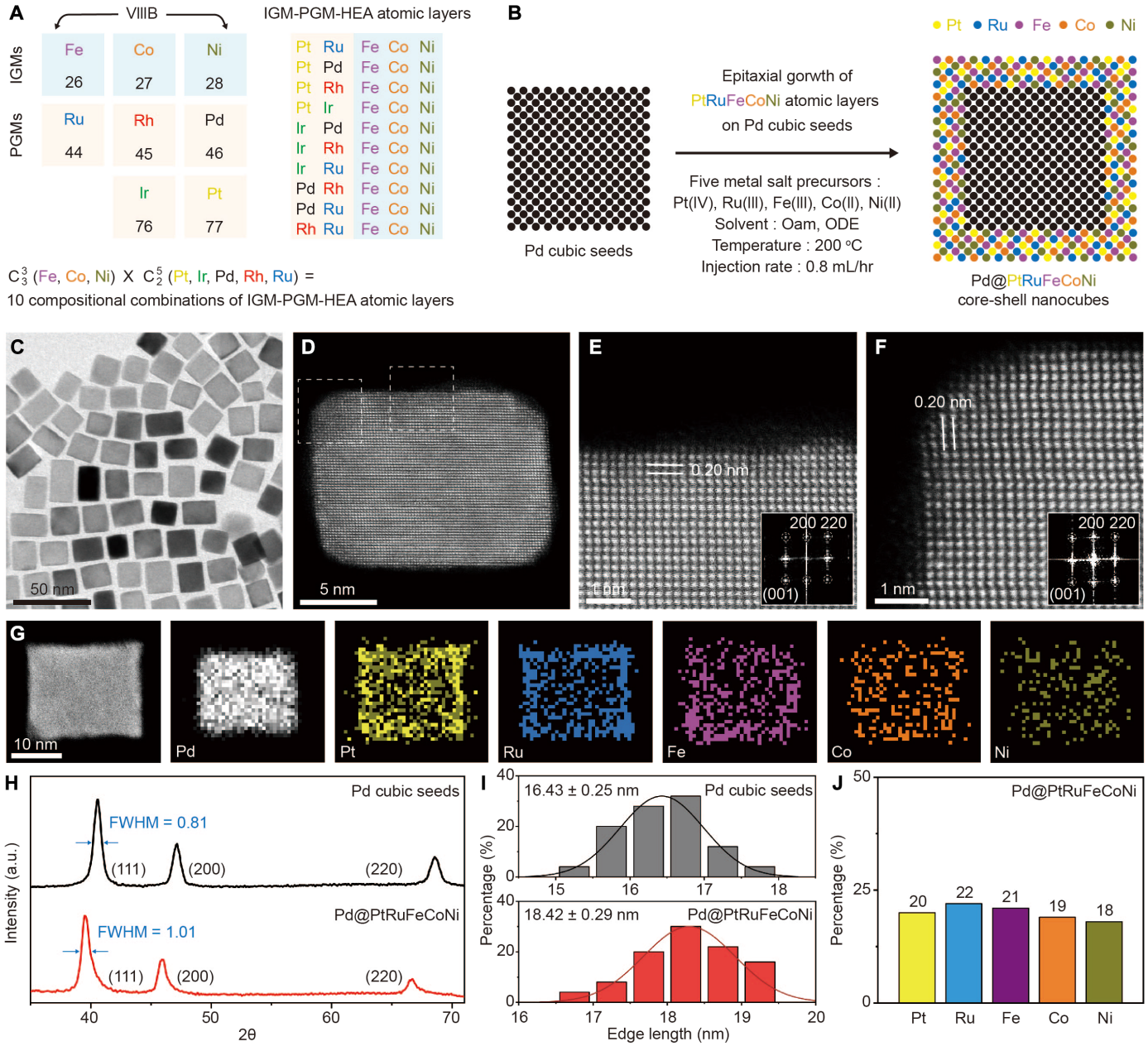


Fig. 1. Synthetic design and structural and compositional characterizations of IGM-PGM-HEA atomic layers. (**A**) Overview of IGMs (highlighted in blue) and PGMs (highlighted in orange) that can form IGM-PGM-HEA atomic layers. (**B**) Schematic of epitaxial growth to obtain PtRuFeCoNi atomic layers on Pd cubic seeds enclosed by {100} facets. (**C** to **G**) TEM, HAADF-STEM, FFT, and EDS mapping analysis of Pd@PtRuFeCoNi core-shell nanocubes. (**H**) XRD analysis and (**I**) size distributions of Pd cubic seeds and Pd@PtRuFeCoNi core-shell nanocubes. (**J**) ICP-OES analysis of all the elements in the PtRuFeCoNi atomic layers. a.u., arbitrary unit.

and EDS elemental analysis. The observed difference can potentially be attributed to two primary reasons. First, the effectiveness of EDS detection tends to diminish for elements with atomic numbers below 30 ($Z < 30$), including Fe ($Z = 26$), Co ($Z = 27$), and Ni ($Z = 28$) (*34*). Second, the potential spatial distributions of the constituent elements may arise within the atomic layers of HEA, which may affect the intensity of EDS signals. Thus, we perform DFT simulations to investigate the spatial distributions within the HEA atomic layers. From a thermodynamics perspective, Fe, Co, and Ni atoms tend to be located at the HEA/Pd interface, demonstrating a comparatively lower energy state (fig. S9), which could support this assumption.

The surface chemical states of Pd@PtRuFeCoNi core-shell nanocubes are studied by x-ray photoelectron spectroscopy (XPS), as shown in Fig. 2 (A to F) and tables S4 and S5. The coexistence of metal and oxidation states is identified by the XPS spectra with curve fittings for all elements, which can be attributed to the spontaneous oxidation on the outermost surface of IGM-PGM-HEA when exposed to air (*9*, *11*, *15*). Compared to the Pt (Fig. 2B), the Ru, Fe,

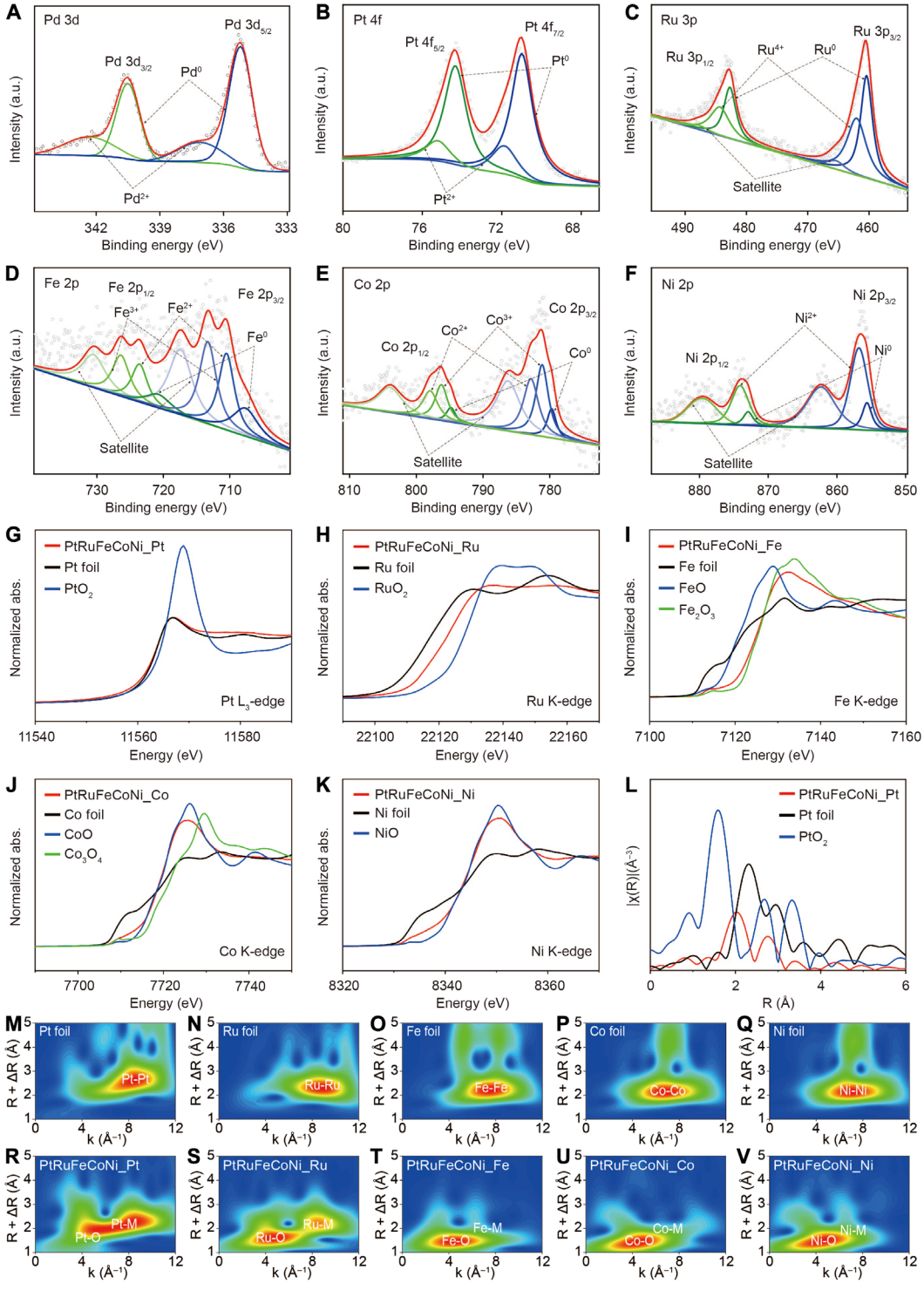


Fig. 2. Surface chemical states and atomic coordination environments of IGM-PGM-HEA atomic layers. (**A** to **F**) XPS spectra of Pd@PtRuFeCoNi core-shell nanocubes: (A) Pd 3d, (B) Pt 4f, (C) Ru 3p, (D) Fe 2p, (E) Co 2p, and (F) Ni 2p. (**G** to **K**) XANES spectra of PtRuFeCoNi atomic layers and their corresponding metallic foils and oxidation states at the (G) Pt L_3-edge, (H) Ru K-edge, (I) Fe K-edge, (J) Co K-edge, and (K) Ni K-edge. (**L**) FT-EXAFS spectra of Pt L_3-edge. (**M** to **V**) Wavelet analysis of PtRuFeCoNi atomic layers and the corresponding metallic foils at the [(M) and (R)] Pt L_3-edge, [(N) and (S)] Ru K-edge, [(O) and (T)] Fe K-edge, [(P) and (U)] Co K-edge, and [(Q) and (V)] Ni K-edge.

Co, and Ni are more easily oxidized, resulting in more substantial amounts of Ru^{+4} (Fig. 2C), Fe^{3+} and Fe^{2+} (Fig. 2D), Co^{3+} and Co^{2+} (Fig. 2E), and Ni^{2+} (Fig. 2F) oxidation states. XPS signals originating from the IGMs of Fe, Co, and Ni all exhibit relatively lower intensities compared to the signals from the PGMs of Pt and Ru. These observations can be attributed to the lower XPS sensitivity factors of the IGMs compared to those of the PGMs (table S5), along with the potential spatial distributions of the constituent elements that may arise within the atomic layers of PtRuFeCoNi, as indicated by the DFT results (fig. S9).

We further use synchrotron XAS to reveal the coordination environment and atomic mixing in the Pd@PtRuFeCoNi core-shell nanocubes (Fig. 2, G to K). The x-ray absorption near edge structure (XANES) spectra of all elements in the PtRuFeCoNi atomic layers, as well as their standard metallic foils and oxides are systematically studied. The post-edges of all elements exhibit deviations in shape and intensity compared to their foil and oxide counterparts, implying the orbital hybridizations of the mixed elements (*10*, *20*). To precisely determine the radial distance and gain deeper insights into the atomic mixing of the five elements within PtRuFeCoNi atomic layers, we conduct the fittings for their Fourier-transformed extended x-ray absorption fine structure (FT-EXAFS) spectra (figs. S10 and S11). The fitting parameters and the resulting fitting outcomes, such as coordination number (CN), radial distance (R), σ^2 (mean square relative displacements), $S0^2$ (passive electron reduction factor), ΔE (energy shift), and R factor, are summarized in table S6. Please note that, as indicated in table S6, "5d" corresponds to Pt, "4d" signifies Ru, and "3d" represents the combination of Fe, Co, and Ni due to the adjacent atomic number of these three elements, respectively.

On the basis of the fitting results (table S6), the radial distance of Pt within the PtRuFeCoNi atomic layers is around 2.733 Å (Pt-5d), showing a subtle increase compared to the 2.716 Å of Pt-4d and 2.674 Å of Pt-3d. For the Pt foil, the radial distance of Pt-Pt bond in the Pt foil is approximately 2.764 Å, closely approaching the actual value for Pt metal (2.774 Å). The Pt-5d radial distance (2.733 Å) within the atomic layers is notably shorter than that of Pt-Pt (2.764 Å) in the Pt foil. This suggests that the PtRuFeCoNi atomic layers, comprising Pt and elements from 3d, 4d, and 5d orbitals, exhibit shorter radial distances in comparison to the Pt-Pt bond observed in the Pt foil. In addition, the coordination values of Pt-5d, Pt-4d, and Pt-3d are calculated as 2.36, 2.94, and 1.81, respectively. These results intriguingly reveal a random distribution of these elements throughout the atomic layers. In the case of Ru, the radial distance of Ru-5d is around 2.713 Å, which is shorter than the radial distance of Pt foil (2.764 Å). The radial distance of Ru-4d is approximately 2.697 Å, closely resembling the radial distance of its metallic foil (2.673 Å). Furthermore, the radial distance of Ru-3d (2.669 Å) appears comparatively longer than the foils of Fe (2.464 Å), Co (2.495 Å), and Ni (2.478 Å), accompanied by the emergence of oxidative bonds around Ru at 2.040 Å. In addition, the coordination values of Ru-5d, Ru-4d, Ru-3d, and Ru-O are determined as 0.52, 3.46, 1.62, and 1.94, respectively. These observations indicate that the radial distances between Ru and the 3d, 4d, and 5d elements are affected by the coordination environment within PtRuFeCoNi atomic layers.

In the case of IGM atoms (table S6), the radial distances of Fe-5d, Fe-4d, Co-5d, Co-4d, Ni-5d, and Ni-4d are notably longer than their metallic foils, implying the formation of metallic bonds between IGM and PGM atoms (i.e., Pt and Ru in this case). Note that the Fe-3d, Co-3d, and Ni-3d exhibit radial distances at approximately 3.1 Å, coupled with Fe-O, Co-O, and Ni-O at around 2.0 Å, demonstrating the emergence of oxidative bonds (*35*). Overall, Pt predominantly exists in the metallic state, while Ru, Fe, Co, and Ni coexist in both metallic and oxidized forms within the PtRuFeCoNi atomic layer, consistent with the observed XPS results (Fig. 2). To study the chemical states of Ru, Fe, Co, and Ni elements, we mathematically determine absorption edge positions by identifying the second derivative inflection points of the pre-edges for Ru, Fe, Co, and Ni elements (fig. S12). Subsequently, we obtain their chemical states through interpolation, establishing a linear relationship between absorption edge positions and the theoretical chemical states of control samples (fig. S13). These observations once again indicate the coexistence of both metallic and oxidative states. This can be attributed to the spontaneous oxidation occurring on the outermost surface of IGM-PGM-HEA atomic layers, which contain elements such as Ru, Fe, Co, and Ni known for their susceptibility to oxidation when exposed to air (*9*, *11*, *15*).

The wavelet transform of EXAFS (WT-EXAFS) is used to compare PtRuFeCoNi atomic layers and their corresponding foils (Fig. 2, M to V). The PtRuFeCoNi atomic layers demonstrate distinctive coordination environments and orbital hybridizations than their metallic foils. In the case of Pt and Ru, the WT-EXAFS analysis reveals additional signals at lower atomic distances and an altered coordination environment as compared to their corresponding foils. The observation indicates the formation of an alloy composite with low–atomic number IGM or the potential emergence of an oxide phase. For Fe, Co, and Ni, signals located at low atomic distances suggest the oxidation of these elements. These findings can be aligned with the FT-EXAFS fitting details in fig. S11 and table S6. In addition, the wider distributions along the *k* axis compared to their standard foils signify the complex coordination environment and the formation of alloy within the PtRuFeCoNi atomic layers. Together, these above XAS results suggest that the five elements undergo atomic mixing within the PtRuFeCoNi atomic layers. This inference is primarily supported by the comparison with Pt (Fig. 2L and table S6), where the radial distance in the alloyed PtRuFeCoNi atomic layers is shorter than that in its metallic foil. The coordination information of Pt-5d, Pt-4d, and Pt-3d also confirms this difference. Despite the more noticeable oxidation in Ru, Fe, Co, and Ni, the presence of an alloying phase in the PtRuFeCoNi atomic layers is confirmed by the distinct radial distances observed between the dissimilar atoms and their respective metallic foils.

A catalyst family of IGM-PGM-HEA atomic layers

We demonstrate the generality of epitaxial growth by extending it to nine other types of IGM-PGM-HEA atomic layers enclosed by {100} facets on the Pd nanocubes, including the PtIrFeCoNi, PtPdFeCoNi, IrPdFeCoNi, IrRuFeCoNi, PdRhFeCoNi, PdRuFeCoNi, PtRhFeCoNi, RhRuFeCoNi, and IrRhFeCoNi, as shown in Fig. 3 (A to I). From the TEM, HAADF-STEM images, and FFT patterns, these Pd@IGM-PGM-HEA core-shell nanocubes all have a cubic shape and {100} facets, together with ultrathin shells. The corresponding EDS maps, line scan, and ICP-OES analysis confirm that the deposited five IGM-PGM elements with more or less the same molar ratios are distributed throughout the entire shells (Fig. 3, fig. S14, and table S3). In particular, for the Pd@PtIrFeCoNi (Fig. 3A), an obvious contrast difference is observed between the PtIrFeCoNi

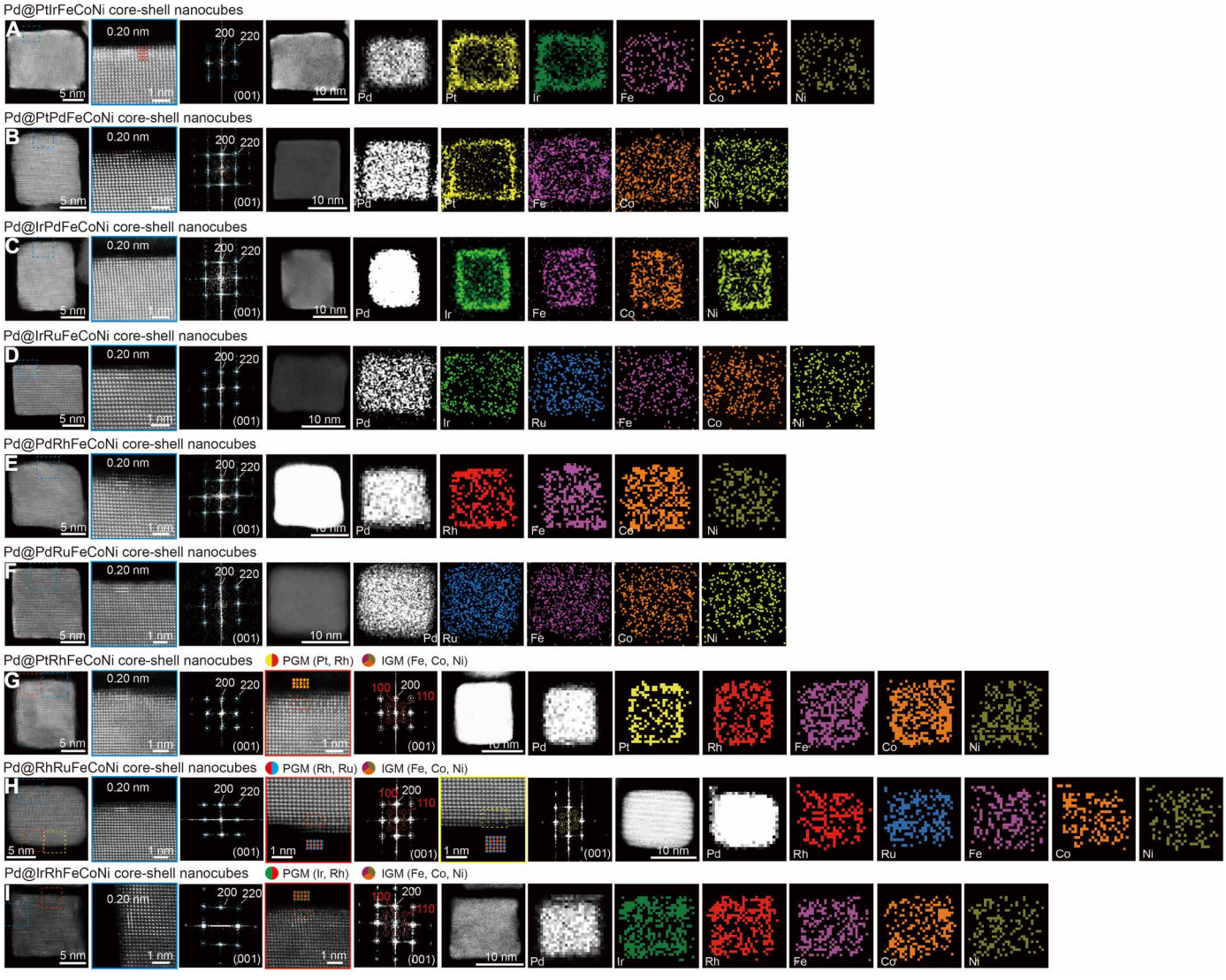


Fig. 3. A catalyst family of IGM-PGM-HEA atomic layers. (**A** to **F**) HAADF-STEM, FFT, and EDS mapping analysis of quinary (A) Pd@PtIrFeCoNi, (B) Pd@PtPdFeCoNi, (C) Pd@IrPdFeCoNi, (D) Pd@IrRuFeCoNi, (E) Pd@PdRhFeCoNi, and (F) Pd@PdRuFeCoNi core-shell nanocrystals. (**G** to **I**) HAADF-STEM, FFT, and EDS mapping analysis of quinary (G) Pd@PtRhFeCoNi, (H) Pd@RhRuFeCoNi, and (I) Pd@IrRhFeCoNi core-shell nanocrystals. The FCC lattices in the local regions of all figures are marked by blue squares. The red squares highlight the $L1_2$-ordered crystal lattices in the local regions in the figures of (G), (H), and (I). The yellow square highlights the $L1_2$-ordered crystal lattice with a dislocation in the figure of (H).

shell (brighter) and the Pd core (darker) in the HAADF-STEM images owing to heavier elements of Pt and Ir in the shell (Fig. 3A). This observation confirms the conformal growth of the PtIrFeCoNi shell with about five atomic layers highlighted by red dots in the atomic-resolution HAADF-STEM image, consistent with the ICP-OES and quantitative analysis (table S3).

For the PtRhFeCoNi (Fig. 3G), RhRuFeCoNi (Fig. 3H), and IrRhFeCoNi (Fig. 3I), the $L1_2$-ordered crystal lattices with the PGMs and IGMs atomic ratio of 3:1 in some local regions of shells highlighted in red squares are also identified by the HAADF-STEM images, in addition to the disordered A1 solid solution (*36*). The corresponding alternating stacking of bright columns of PGM atoms and dark columns of IGM atoms are confirmed by the line profiles (fig. S15), showing the intensity contrasts between the PGM columns and the IGM columns in these three samples with the locally ordered structures. Besides, the FFT patterns within the selected regions marked by red squares further confirm the presence of additional (100) and (110) superlattice spots (Fig. 3, G to I, and figs. S16 and S17), indicated by red dotted circles (*24*, *25*). These results imply that some Fe, Co, and Ni atoms can be randomly distributed on the corner sites of the FCC structure, while two elements of PGMs (i.e., Pt/Rh, Rh/Ru, and Ir/Rh) can occupy the face-centered sites (fig. S16), although the temperature of synthesis in the present study is relatively low compared to the previous reports (*13*, *16*, *24*). In addition, the slight displacement of diffraction points, resulting in a skewed square in the FFT pattern of RhRuFeCoNi atomic layers, is also observed in the yellow square of Fig. 3H. This deviation from the anticipated symmetry of an ideal intermetallic $L1_2$-ordered crystal

lattice suggests the presence of a local dislocation within the intermetallic RhRuFeCoNi atomic layers (fig. S17), as revealed by a series of TEM characterizations.

A metastable phase of IGM-PGM-HEA atomic layers identified by in situ HAADF-STEM

We further perform in situ HAADF-STEM to identify whether PtRuFeCoNi atomic layers on the Pd nanocubes are thermo dynamically stable or kinetically favored state when subjected to heating up to 500°C in the vacuum, as shown in Fig. 4. Again, the atomic-resolution HAADF-STEM images and their corresponding FFT patterns are viewed along the [001] zone axis. The morphology of the cubic nanocrystal is preserved upon heating to 200°C and then gradually changed to a pseudospherical shape when further heated to 500°C. On the other hand, the PtRuFeCoNi atomic layers show a solid-solution FCC structure with randomly distributed mixed atoms below 200°C (Fig. 4, A to O). Further heating to 300°C (Fig. 4, P to T) and 400°C (Fig. 4, U to Y) leads to the local phase transformation of PtRuFeCoNi atomic layers, changing the phase to partial $L1_0$-ordered and $L1_2$-ordered intermetallic phases, respectively (figs. S16, S18, and S19). In addition, the corresponding intensity profiles extracted from the atomic-resolution HAADF-STEM images of the $L1_0$-ordered and $L1_2$-ordered phase structures of PtRuFeCoNi atomic layers (figs. S18 and S19) also validate the local phase transformation of PtRuFeCoNi atomic layers at 300° and 400°C, respectively. Note that the $L1_0$ phase is the tetragonal distortion of the FCC structure with the PGMs and IGMs atomic ratio of 1:1, in which the PGMs occupy one sublattice and the IGMs occupy the other sublattice (fig. S16) (*16*). The presence of additional (110) and (210) superlattice spots in the FFT patterns (Fig. 4, S and T) further confirms the existence of the $L1_0$ phase, highlighted by blue dotted circles (*37*, *38*). Such a local disorder-to-order transition could be attributed to the inward diffusion of IGM atoms from the shell into the Pd core due to lower diffusion barriers of IGMs with smaller radii (*24*), further resulting in the vacant sites of IGMs filled by the outward-diffusing PGM atoms from the shell or Pd atoms from the core.

Elevating the temperature further to 500°C leads to phase separation, with clear phase boundaries shown in the HAADF-STEM image (Fig. 4Z). In particular, the Pt skin formed from the decomposition of the formed intermetallic phase at the surface can be directly observed in the HAADF-STEM image (*24*) (Fig. 4AA) owing to the highest Z contrast of Pt in the nanocrystals. From the FFT pattern taken from this surface region (Fig. 4AC), the nanocube has three sets of diffraction patterns, including the FCC polycrystalline Pt, single-crystal Pd, and $L1_2$ intermetallic phases (fig. S20). The HAADF-STEM image (Fig. 4AB) and FFT pattern taken from the core region (Fig. 4AD) only show the coexistence of Pd and $L1_2$ lattices. These observations suggest the interdiffusion of IGMs and PGMs for controlling the ordering transformation (Fig. 4AE). We reveal the transformation kinetics of phase from high-entropy mixing (25°C), to ordered $L1_0$ (300°C) and $L1_2$ (400°C) intermetallic, and to phase-separated phases (500°C), indicating the metastable phase of the far-from-equilibrium PtRuFeCoNi atomic layers (Fig. 4AF). This post-annealing treatment provides a promising avenue for achieving facet control in high-entropy intermetallic atomic layers derived from multicomponent solid solutions. This presents an important advancement, particularly considering the challenges encountered with conventional synthetic methods.

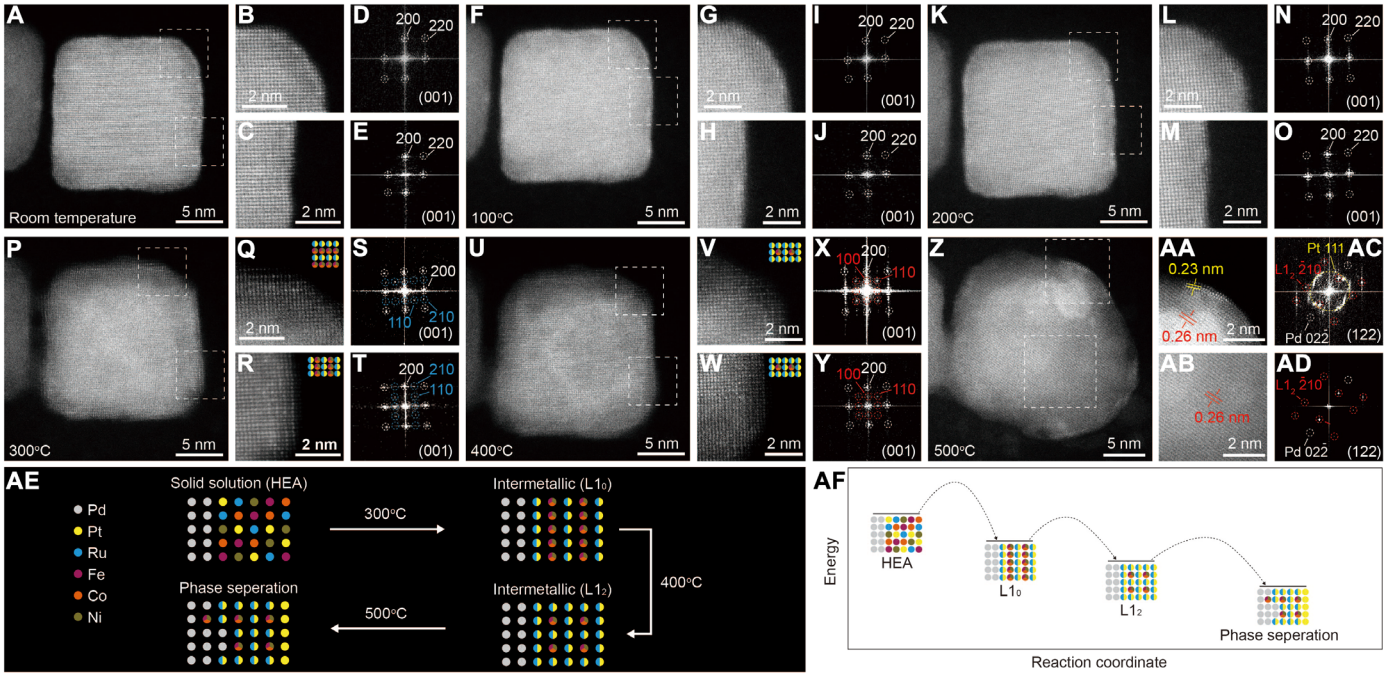


Fig. 4. A metastable phase of IGM-PGM-HEA atomic layers identified by in situ HAADF-STEM. HAADF-STEM images and their corresponding FFT patterns of Pd@PtRuFeCoNi core-shell nanocubes at different temperatures: (**A** to **E**) room temperature, (**F** to **J**) 100°C, (**K** to **O**) 200°C, (**P** to **T**) 300°C, (**U** to **Y**) 400°C, and (**Z** to **AD**) 500°C. (**AE**) Schematic illustration showing the phase transformation from solid-solution PtRuFeCoNi atomic layers to ordered $L1_0$ and $L1_2$ intermetallic and phase-separated states. (**AF**) Schematic illustration showing PtRuFeCoNi atomic layers as a kinetically favored phase.

Electrocatalytic performance and mechanism

To evaluate the potential applications of a catalyst family of IGM-PGM-HEA atomic layers, we select 10 different IGM-PGM-HEA covered by {100} facets on the Pd nanocubes as catalysts for electrocatalytic HER and HOR. In the first set of catalytic measurements, we evaluate the catalytic performance of various catalysts for HER in an acidic 0.5 M H_2SO_4 electrolyte, as shown in Fig. 5 (A to D). The corresponding catalysts, including commercial Pt/C, mono-, bi-, and trimetallic samples, are used as references. As confirmed by the linear sweep voltammetry (LSV) (fig. S21), the PtRuFeCoNi atomic layers exhibit the lowest HER overpotential of 41.3 mV (versus the reversible hydrogen electrode, RHE) at a geometric current density of −10 mA cm^{-2} among these catalysts. Figure 5A shows the intrinsic activities normalized to the electrochemical surface area (ECSA) of the catalysts (fig. S22) to characterize their intrinsic activities. The PtRuFeCoNi atomic layers reach the highest intrinsic activity of 21.95 mA cm^{-2} at −0.1 V, which is 4.5 times greater than that of the commercial Pt/C catalysts (4.93 mA cm^{-2}), as shown in Fig. 5B, confirming the merit of high-entropy electrocatalysts consisting of IGMs and PGMs: higher catalytic activity and diminished usage of noble metals. In addition, we further manipulate the composition of the PtRuFeCoNi atomic layers covered by the {100} facets, leading to the two distinct atomic layers with low and high compositions of PGMs (fig. S23). The elemental ratios in the low-PGM atomic layers are 11:11:24:20:22 for Pt/Ru/Fe/Co/Ni, while the elemental ratios in the high-PGM atomic layers are 28:30:12:15:15 for Pt/Ru/Fe/Co/Ni, respectively. The catalytic activity of PtRuFeCoNi atomic layers with a moderate level of PGM content (42%) (Fig. 5B) surpasses those of atomic layers with low-PGM content (22%) and high-PGM content (58%) (fig. S23). These results suggest that PtRuFeCoNi atomic layers, despite the potential presence of partial inhomogeneities within them as indicated by DFT simulations (fig. S9), function as optimal HER catalysts, owing to their nearly equal elemental ratios (Fig. 5A). Furthermore, the durability of the PtRuFeCoNi atomic layers is improved compared to the commercial Pt/C catalysts. After 15,000 cycles of testing (Fig. 5, C and D, and fig. S24), the Pd@PtRuFeCoNi catalysts only display a 7.3% drop in the intrinsic activity, as opposed to a 22.3% drop for the intrinsic activity of the commercial Pt/C catalysts.

In the second set of catalytic measurements, as shown in Fig. 5 (E to H), we study the catalytic performance of the catalysts for alkaline HOR, an important process for hydrogen fuel cells. Their HOR LSV polarization curves measured in H_2-saturated 0.1 M KOH solution are shown in Fig. 5E. The Pd@PtRuFeCoNi core-shell nanocubes exhibit the lowest HOR overpotential of 86 mV at an intrinsic current density of 0.5 mA cm^{-2} among all the catalysts. As shown in Fig. 5F, the intrinsic activity of Pd@PtRuFeCoNi core-shell nanocubes is 0.82 mA cm^{-2} at 0.1 V, which is 5.5 times higher than that of the commercial Pt/C catalysts. Moreover, the durability is evaluated by measuring the HOR activities of Pd@PtRuFeCoNi and commercial Pt/C catalysts (Fig. 5, G and H, and fig. S25). Only a 16.0% loss in intrinsic activity is observed after 3000 cycles of testing for the Pd@PtRuFeCoNi catalysts. In contrast, the intrinsic activity dropped by 55.1% for the commercial Pt/C catalysts after 3000 cycles. These results demonstrate that the metastable phase of PtRuFeCoNi atomic layers with square atomic arrangements can serve as a highly active catalyst with remarkable long-term durability for both HER and HOR.

Over the past decade, metastable phase catalysts have shown great promise in optimizing catalytic activity, leveraging the strong association between electronic structure and crystal configuration (*39*). These metastable catalysts, when appropriately chosen for specific reaction conditions, also exhibit excellent long-term durability across various catalytic reactions. Taking the example of our metastable PtRuFeCoNi atomic layers (i.e., kinetically controlled products), as mentioned above, they demonstrate exceptional activity and long-term durability under the relatively low-temperature conditions typical of electrocatalytic HER and HOR reactions (Figs. 5, 6, and 7). It should be emphasized that electrocatalysts used in the processes of HER and HOR, essential for water splitting and fuel cells, respectively, typically operate under conditions with temperatures below 80°C (*40*–*42*). This phenomenon can be attributed to the slow atomic diffusion at low temperatures, which may constrain the structure to kinetically favored states. Given that the kinetical products could be easily trapped in numerous states corresponding to local minima in Gibbs free energy (*43*), this limitation impedes the transformation into thermodynamically controlled products, thereby contributing to the observed durability.

Furthermore, we performed operando XAS measurements to identify the actual active sites within the HEA atomic layers for both HER (Fig. 6) and HOR (Fig. 7). It is worth highlighting that operando XAS measurements serve as powerful tools for unraveling the catalytic mechanism. These measurements provide insights into the interactions among elements within a high-entropy multielemental phase, as well as the dynamic interplay between the actual active sites and the reactants, intermediates, or products during the catalysis (*10*, *44*, *45*). During the acidic HER, the XANES spectra (Fig. 6, A to E) of Pt L_3-edge, Ru K-edge, Fe K-edge, Co K-edge, and Ni K-edge and the corresponding FT-EXAFS spectra (Fig. 6, F to J) are collected at applied potentials from 0.05, −0.05 to −0.15 V versus RHE. When the cathodic potential of −0.05 V versus RHE is applied, the absorption edge of Fe shifts to higher energy, while the Ru absorption edge moves to lower energy. This interaction could be attributed to the electron transfer from Fe (Fig. 6C) to Ru (Fig. 6B) sites during HER, which enhances the affinity of Ru to the attached hydrogen ions and thus accelerate hydrogen production (*10*, *46*, *47*). Note that there are no substantial changes in the absorption edges of Pt (Fig. 6A), Co (Fig. 6D), and Ni (Fig. 6E) elements with the increased applied voltage, suggesting that Ru atoms serve as direct active sites on the PtRuFeCoNi atomic layers during the HER. Moreover, the white-line height of Ru waxes and wanes during HER, meaning that the Ru orbitals are involved in the catalytic HER process. Our control experiment (Fig. 5B) also demonstrates that the PtRuFeCoNi atomic layers (21.95 mA cm^{-2}) also show a higher intrinsic activity compared with the bimetallic RuFe atomic layers (12.88 mA cm^{-2}). Such comparison implies that these nonactive Pt, Co, and Ni elements mixed in PtRuFeCoNi can engage in the orbital hybridizations of the mixed elements confirmed by XAS (Fig. 2, G to K) and increase the entropy of mixing, which could make up the solid-solution alloy work in synergy with each other to achieve better catalytic performance. Furthermore, the FT-EXAFS of Pt, Ru, Fe, Co, and Ni show no identifiable changes when applying a potential from 0.05 to −0.15 V_{RHE}, indicating that the PtRuFeCoNi atomic layers have exceptional durability (Fig. 6, F to J, and fig. S26).

Furthermore, we extended our investigation to identify the specific active sites within the PtRuFeCoNi atomic layers for HOR under H_2-saturated conditions in 0.1 M KOH electrolyte, using operando XAS measurements again (Fig. 7). Typically, the HOR process involves the adsorption of OH species (OH_{ad}), along with the dissociation of

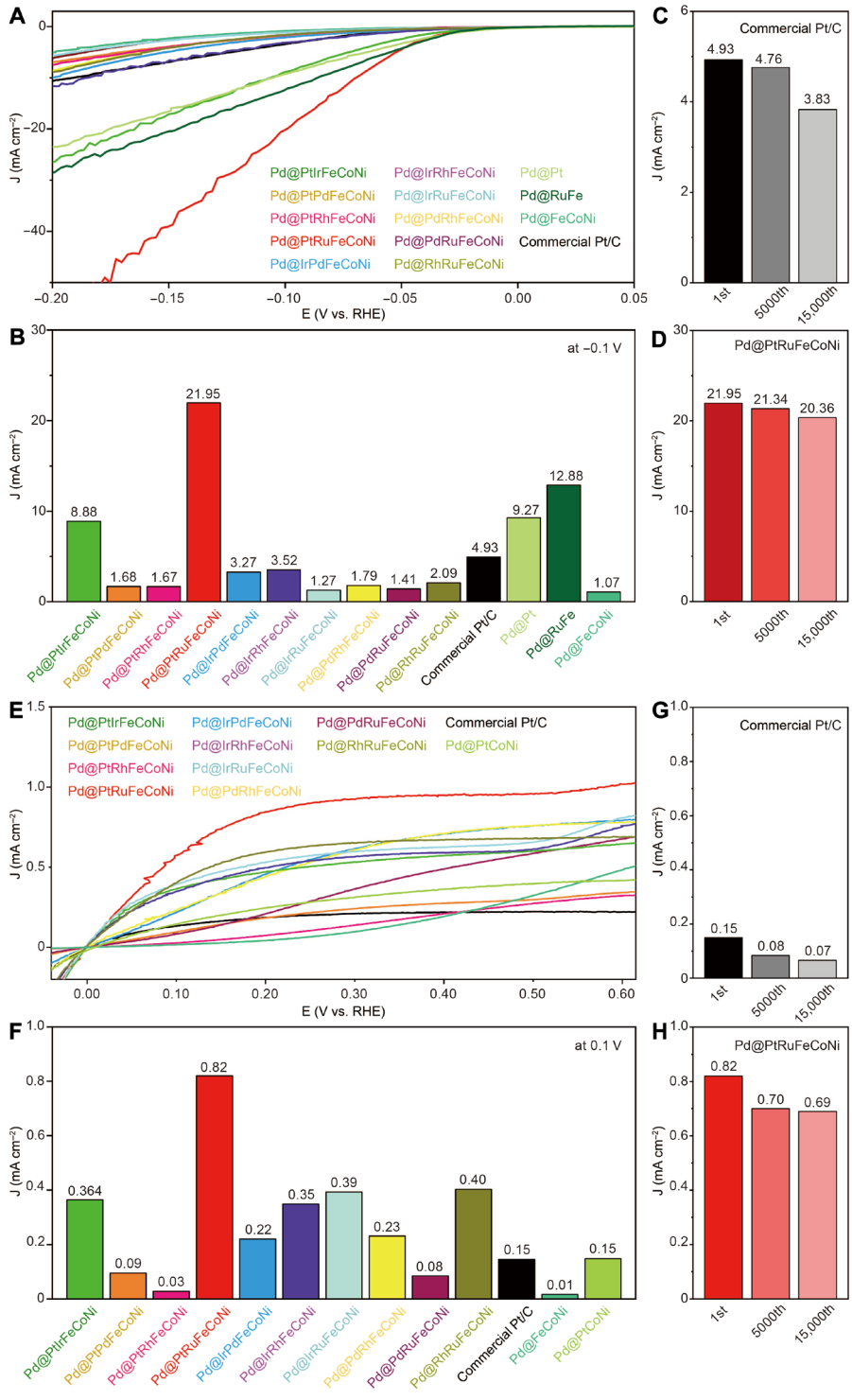


Fig. 5. Catalytic performance of IGM-PGM-HEA atomic layers. (**A** to **D**) Electrocatalytic HER in the 0.5 M H_2SO_4. (A) Polarization curves, (B) comparison of intrinsic activities at −0.1 V, and durability tests of (C) commercial Pt/C, and (D) Pd@PtRuFeCoNi core-shell nanocubes. (**E** to **H**) Electrocatalytic HOR in the 0.1 M KOH. (E) Polarization curves, (F) comparison of intrinsic activities at 0.1 V, and durability tests of (G) commercial Pt/C, and (H) Pd@PtRuFeCoNi core-shell nanocubes.

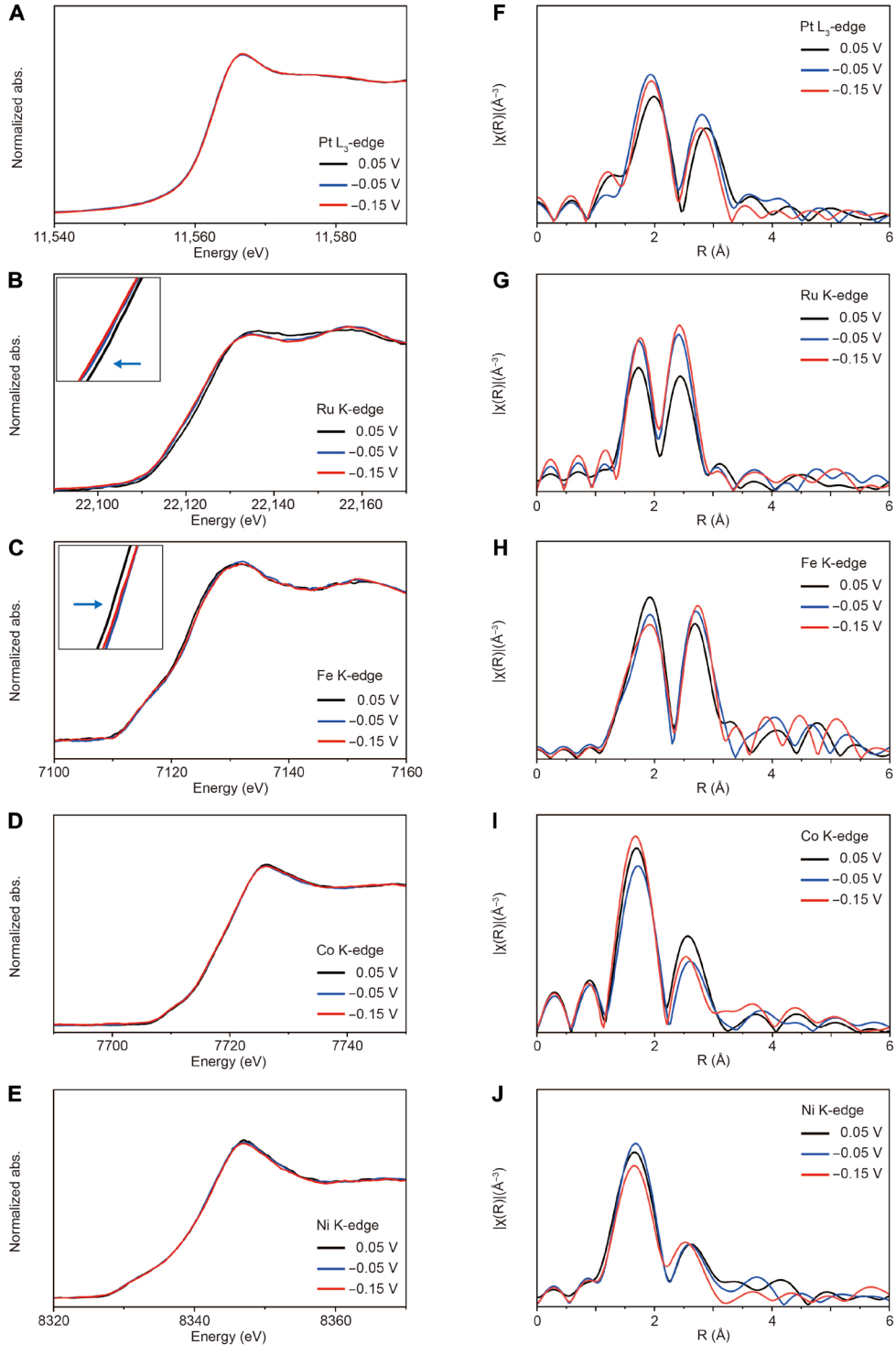


Fig. 6. Catalytic mechanism of HER revealed by operando XAS during catalysis. (**A** to **E**) Operando XANES spectra of PtRuFeCoNi atomic layers under the applied potentials from 0.05, −0.05 to −0.15 V versus RHE in 0.5 M H_2SO_4 during HER at the (A) Pt L_3-edge, (B) Ru K-edge, (C) Fe K-edge, (D) Co K-edge, and (E) Ni K-edge. (**F** to **J**) Operando FT-EXAFS spectra of PtRuFeCoNi atomic layers under the applied potentials from 0.05, −0.05 to −0.15 V versus RHE during HER at the (F) Pt L_3-edge, (G) Ru K-edge, (H) Fe K-edge, (I) Co K-edge, and (J) Ni K-edge.

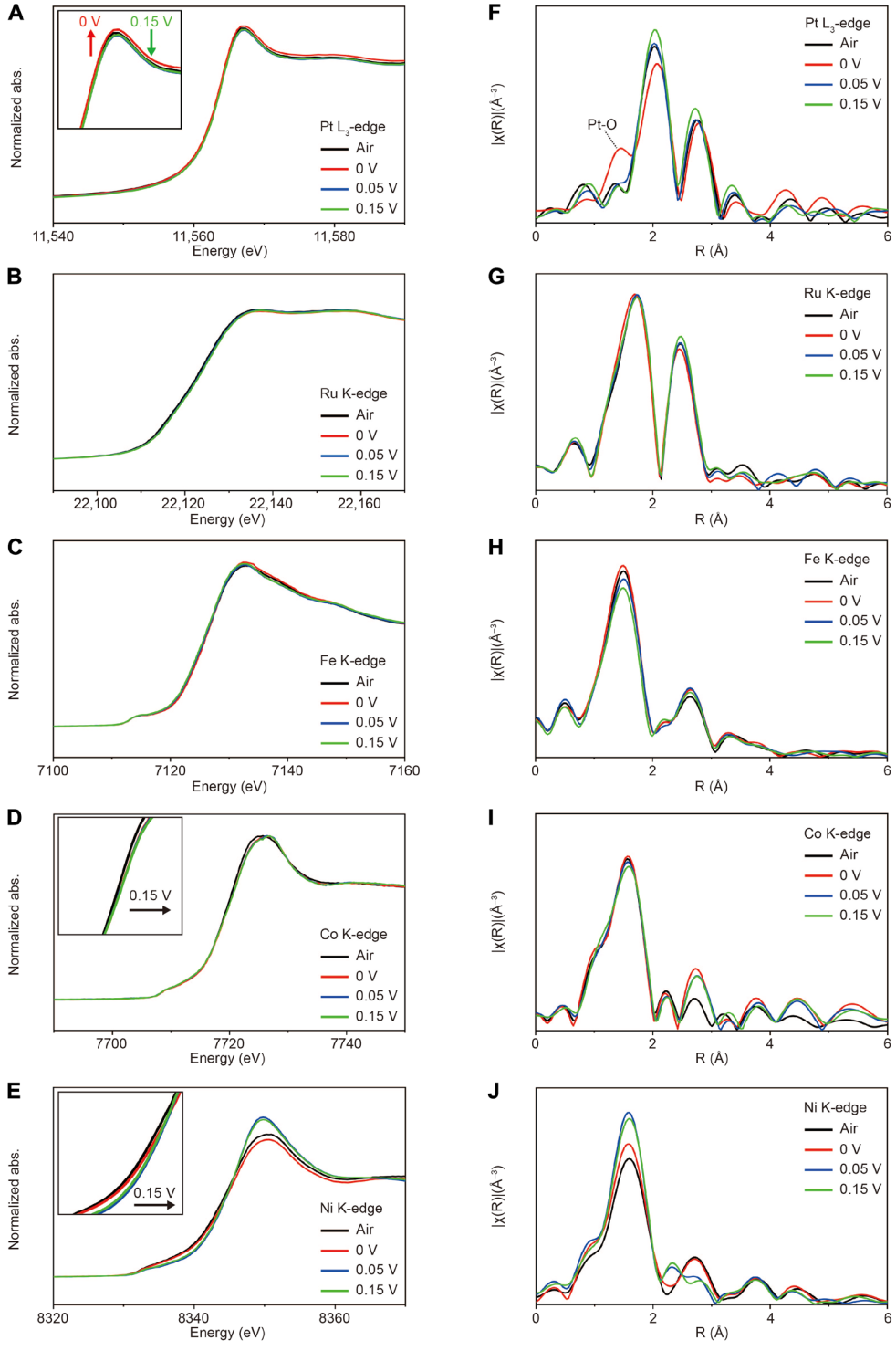


Fig. 7. Catalytic mechanism of HOR revealed by operando XAS during catalysis. (**A** to **E**) Operando XANES spectra of PtRuFeCoNi atomic layers, including measurements in air and applied potentials ranging from 0, 0.05, to 0.15 V versus RHE in 0.1 M KOH during HOR at the (A) Pt L_3-edge, (B) Ru K-edge, (C) Fe K-edge, (D) Co K-edge, and (E) Ni K-edge. (**F** to **J**) Operando FT-EXAFS spectra of PtRuFeCoNi atomic layers, including measurements in air and applied potentials ranging from 0, 0.05, to 0.15 V versus RHE during HOR at the (F) Pt L_3-edge, (G) Ru K-edge, (H) Fe K-edge, (I) Co K-edge, and (J) Ni K-edge.

molecular H_2 and subsequent adsorption (*44*, *45*). The XANES spectra (Fig. 7, A to E) of Pt L_3-edge, Ru K-edge, Fe K-edge, Co K-edge, and Ni K-edge and their corresponding FT-EXAFS spectra (Fig. 7, F to J) are collected in the air and under the applied potentials from 0, 0.05, to 0.15 V versus RHE in the electrolyte for the PtRuFeCoNi atomic layers. At the Pt L_3-edge, the white-line intensity in the XANES spectra increases when PtRuFeCoNi atomic layers are immersed in the alkaline electrolyte at 0 V. Subsequently, it decreases with the rise in applied potentials from 0, 0.05, to 0.15 V (Fig. 7A). Furthermore, the corresponding FT-EXAFS data collected at 0 V reveal a distinct peak associated with Pt-O at approximately 1.5 Å, which gradually diminishes as the applied voltage increases. These observations suggest the initial OH_{ad} on the Pt sites in the PtRuFeCoNi atomic layers at 0 V, followed by the desorption process as the applied voltage increases (Fig. 7F) (*44*, *45*). At the Co K-edge, the absorption edge experiences a shift to higher energy at 0 V in the alkaline solution and then remains stable with increasing applied potentials to 0.05 and 0.15 V (Fig. 7D) (*44*, *45*, *48*, *49*). In addition, a subtle shoulder emerges around the Co-O peak in the FT-EXAFS spectra when applying the potentials. These findings suggest that Co sites may also be actively involved in the adsorption process of OH species (Fig. 7I). In addition, the energy positions of the absorption edges for the Ni K-edge in the XANES spectra gradually shift to higher energy with increasing applied voltage, indicating an elevated chemical state for Ni at relatively higher applied potentials (Fig. 7E and fig. S27). In addition, the peak in the Ni K-edge FT-EXAFS spectra around 1.7 Å also increases as the applied voltage rises. These changes imply the involvement of the OH_{ad} on the Ni sites (Fig. 7J) (*44*, *45*, *48*, *49*).

In contrast, the XANES and FT-EXAFS spectra of the Ru K-edge and Fe K-edge exhibit negligible changes with increasing applied potentials (Fig. 7, B to C and G to H). These observations suggest that Pt, Co, and Ni atoms act as "*actual active sites*" for interacting with OH^- species during HOR. The hydroxyl binding energy theory (*48*, *49*), developed to explain the catalytic mechanism of HOR, underscores the significance of the formation of adsorbed hydroxide (i.e., OH_{ad}) species on the surface—a crucial factor influencing catalytic activity for HOR. Furthermore, our control experiment (Fig. 5F and fig. S28) highlights that the PtRuFeCoNi atomic layers (0.82 mA cm^{-2}) exhibit a higher intrinsic activity when compared to the trimetallic PtCoNi atomic layers (0.15 mA cm^{-2}) at 0.1 V. This comparison suggests that the inclusion of nonactive Ru and Fe elements in the PtRuFeCoNi atomic layers can engage in orbital hybridizations of PtRuFeCoNi atomic layers (Fig. 2, G to K) and enhance the entropy of mixing. These interactions enable the solid-solution alloy to function synergistically, resulting in improved catalytic performance.

At this stage, the reasons contributing to the exceptional catalytic performance of Pd@PtRuFeCoNi remain uncertain. Therefore, we have used additional DFT simulations to delve into the underlying reasons, focusing particularly on the HER for investigation (Fig. 8). Notably, when Pt is substituted with Ir in Pd@IrRuFeCoNi, there is a substantial decline in HER activity, amounting to only approximately 6% of the activity observed in Pd@PtRuFeCoNi (Fig. 5B). To understand the origin of this substantial difference, we computed hydrogen-adsorption free energy (ΔG_{H^*}, which is widely used descriptor for the ability of HER) for these two HEA nanocubes using DFT. To consider the diverse chemical configurations of an HEA surface, we sampled 117 H-adsorption configurations for both Pd@PtRuFeCoNi and Pd@IrRuFeCoNi, resulting in approximately 50

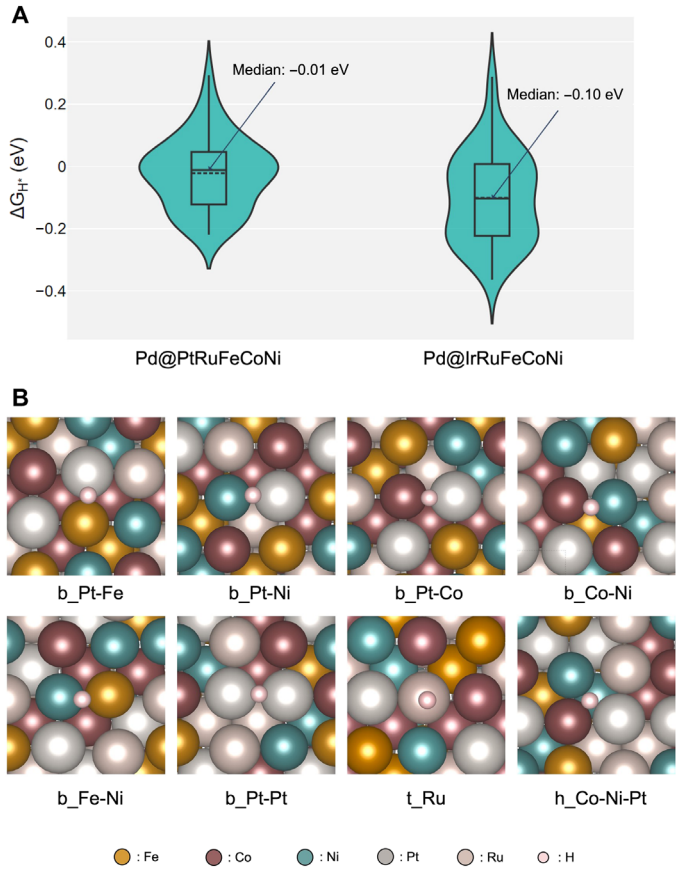


Fig. 8. Hydrogen adsorption behaviors on Pd@PtRuFeCoNi and Pd@IrRuFeCoNi investigated using DFT. (**A**) The distribution of hydrogen-adsorption free energy ΔG_{H^*}. (**B**) The H-adsorption configurations of Pd@PtRuFeCoNi with an ideal H-adsorption strength for HER (we set $|\Delta G_{H^*}| < 0.02$ eV). The prefix "b," "t," and "h" represent the bridge, on-top, and hollow adsorption sites, respectively.

unique H-adsorption configurations after geometry optimization. From the distribution of ΔG_{H^*} shown in Fig. 8A, the Pd@PtRuFeCoNi exhibit a closer alignment of median ΔG_{H^*} values (−0.01 eV) to the ideal (0 eV) in comparison to Pd@IrRuFeCoNi (−0.10 eV). The relative HER activity at room temperature ($k_{IrRuFeCoNi}/k_{PtRuFeCoNi}$) of these two materials resulting from the different ΔG_{H^*} could be roughly estimated using an Arrhenius-type equation (*50*), which gives

$$\frac{k_{IrRuFeCoNi}}{k_{PtRuFeCoNi}} = \exp\left[-\frac{(-0.01+0.10)}{0.0257}\right] \approx 3\% \quad (1)$$

This agrees reasonably well with our experimental observation (~6%). In addition to HER taking place on the HEA surface, we also examine the potential hydrogen spillover process from HEA to the carbon support that could introduce extra HER activity. Recent work by Li *et al.* (*51*) suggests that the work function of the metal is a crucial factor in the hydrogen spillover process. Our DFT results reveal that the work function values of Pd@PtRuFeCoNi and Pd@IrRuFeCoNi are highly comparable (4.53 ± 0.05 versus 4.55 ± 0.02 eV). This suggests that the hydrogen spillover process is not the determining

factor for the distinct HER activity observed between these two materials.

Next, to understand the origin of the more negative ΔG_{H^*} in Pd@IrRuFeCoNi compared to Pd@PtRuFeCoNi, we designed a (100) surface model of Pt and Ir subject to Pd lattice constant, simulating the nanocubes of Pd@Pt and Pd@Ir, respectively. Our investigation revealed that the ΔG_{H^*} of Pd@Ir is consistently more negative than that of Pd@Pt across all H-adsorption sites, as detailed in table S7. This observation suggests a superior H-binding strength of Ir over Pt. Consequently, the more negative ΔG_{H^*} distribution observed in Pd@IrRuFeCoNi compared to Pd@PtRuFeCoNi can be attributed to the enhanced H-binding affinity of Ir, providing insights into the distinct reactivity patterns between the two configurations.

Last, to elucidate the outstanding HER activity of Pd@PtRuFeCoNi, we focused on exploring H-adsorption configurations with ΔG_{H^*} values closely aligned with the ideal one. Notably, most of these optimal adsorption sites were identified as bridge sites, as depicted in Fig. 8B. Furthermore, it was observed that the H-adsorption strength consistently diminished across various metal elements (M) when comparing Pd@PtRuFeCoNi to Pd@M (fig. S29). For instance, the ΔG_{H^*} for on-top adsorption on Ru decreased from −0.10 eV (Pd@Ru, table S7) to −0.01 eV (Pd@PtRuFeCoNi, t-Ru in Fig. 8B). Another example is the bridge adsorption between Pt and Co, which exhibits a ΔG_{H^*} of 0.01 eV for Pd@PtRuFeCoNi (b_Pt-Co in Fig. 8B). Unexpectedly, when considering that the H-adsorption strength of b_Pt-Co is an average of b_Pt-Pt and b_Co-Co, the ΔG_{H^*} of b_Pt-Co was expected to fall between those of b_Pt-Pt and b_Co-Co. However, both b_Pt-Pt and b_Co-Co displayed more negative ΔG_{H^*} values, namely −0.24 and − 0.12 eV, respectively (table S7). This suggests that the cocktail effect contributes to the exceptional HER activity of Pd@PtRuFeCoNi (*1*, *3*, *10*, *11*, *20*). It is crucial to underscore that the cocktail effect observed in HEA structures is widely recognized as a complex and synergistic mechanism that plays a pivotal role in yielding exceptional catalytic performance. The collaborative interactions among the diverse multielement compositions lead to substantial deviations in properties when compared to single-element metals. Notably, the intriguing aspect lies in the fact that the intricate workings of such multicomponent synergy within HEA structures remain largely elusive in this field. Conducting further investigations into the synergistic mechanisms, particularly from the perspectives of local lattice distortion and electronic interaction within HEA structures, using advanced techniques such as high-throughput computations and machine-learning tools, holds the potential to advance our understanding of the cocktail effect on catalytic performance (*3*).

DISCUSSION

This work reports a catalyst family of IGM-PGM-HEA–based nanocrystals with controlled facets by wet chemical synthesis. Ten different compositions of IGM-PGM-HEA atomic layers with all the square atomic arrangements are synthesized and evaluated for their HER and HOR catalytic performances. The atomic layers containing Pt, Ru, Fe, Co, and Ni exhibit superior catalytic activity and durability for both HER and HOR when compared to other IGM-PGM-HEA and commercial Pt/C catalysts. We elaborate on the corresponding atomic-scale synergistic interactions, active sites, and catalytic mechanisms during catalysis through operando synchrotron XAS measurements and DFT simulations. We reveal the cocktail effect arising from the multielement composition, which can optimize hydrogen-adsorption free energy ($\Delta G_{H^*} = -0.01$ eV) and plays a crucial role in the remarkable catalytic activity observed in PtRuFeCoNi atomic layers. Furthermore, in situ HAADF-STEM confirms the metastable phase of PtRuFeCoNi atomic layers. It captures the key phase transformation from disordered atomic mixing to ordered $L1_0$ and $L1_2$ intermetallic structures and phase-separated lattices under thermal stress. This work not only opens a paradigm for developing the facet-controlled IGM-PGM-HEA nanocrystals but also promotes fundamental insights into their catalytic mechanism and thermal stability.

MATERIALS AND METHODS

Materials

Palladium acetylacetonate [$Pd(C_5H_7O_2)_2$], sodium tetrachloropalladate (Na_2PdCl_4), iron acetylacetonate [$Fe(C_5H_7O_2)_3$], nickel acetylacetonate [$Ni(C_5H_7O_2)_2$], cobalt acetylacetonate [$Co(C_5H_7O_2)_2$], dihydrogen hexachloroiridate ($H_2IrCl_6{\cdot}xH_2O$), chloroplatinic acid hexahydrate ($H_2PtCl_6{\cdot}6H_2O$), ruthenium chloride hydrate ($RuCl_3{\cdot}xH_2O$), rhodium chloride hydrate ($RhCl_3{\cdot}xH_2O$), L-ascorbic acid (AA), poly(vinyl pyrrolidone) (PVP, $M_W \approx 55{,}000$), Oam, ODE, formaldehyde (HCHO), *n*-hexane [$CH_3(CH_2)_4CH_3$], and potassium bromide (KBr) are all purchased from Sigma-Aldrich and used after received. Ethylene glycol (EG) is purchased from J. T. Baker. Deionized (DI) water (18.2 MΩ·cm) is used for all experiments.

Synthesis of 16.8-nm Pd cubic seeds

To synthesize the 16.8-nm Pd cubic seeds, a standardized procedure is followed (*52*). An 8-ml aqueous solution containing AA (60 mg), PVP (105 mg), and KBr (600 mg) is added to a 23-ml vial. Subsequently, the solution undergoes preheating at 80°C for 10 min. Following the preheating step, a one-shot injection of 3-ml Na_2PdCl_4 aqueous solution with a concentration of 19 mg/ml is added into the vial in one shot. The reaction mixture is capped and kept at 80°C for 3 hours under magnetic stirring and then cooled to room temperature. The Pd cubic seeds are dispersed in Oam solvent after a cleaning procedure involving centrifugation and washing with DI water three times.

Synthesis of 18.1-nm Pd octahedral seeds

First, we prepare 12-nm Pd nanocubes as the seeds to obtain the seed-mediated growth of 18.1-nm Pd octahedral seeds (*52*). To synthesize the 12-nm Pd nanocubes, an 8 ml aqueous solution containing AA (60 mg), PVP (105 mg), and KBr (300 mg) is introduced into a 23-ml vial. The solution is subsequently preheated at 80°C for 10 min. Following the preheating step, a one-shot injection of 3-ml Na_2PdCl_4 aqueous solution with a concentration of 19 mg/ml is added into the vial in one shot. The reaction mixture is capped and kept at 80°C for 3 hour under magnetic stirring and then cooled to room temperature. After the reaction, the Pd nanocubes are dispersed in EG solvent after a cleaning procedure including centrifugation and washing with DI water three times. Second, a typical procedure is used to synthesize the 18.1-nm Pd octahedral seeds. Three-hundred microliters of the 12-nm Pd nanocubes with a concentration of 1.8 mg ml^{-1} are mixed with 8 ml of an aqueous mixture containing 100 μl of HCHO and PVP (105 mg) in a 23-ml vial. The solution is subsequently preheated at 60°C for 10 min. Following the preheating step, a one-shot injection of 3 ml Na_2PdCl_4 aqueous solution with a concentration of 12.6 mg/ml is added into

the vial in one shot. The reaction mixture is capped and kept at 60°C for 3 hours under magnetic stirring and then cooled to room temperature. After the reaction, the Pd octahedral seeds are dispersed in Oam after a cleaning procedure involving centrifugation and washing with DI water three times.

Synthesis of IGM-PGM-HEA atomic layers on the Pd seeds using epitaxial growth

Through the epitaxial growth, we synthesize IGM-PGM-HEA atomic layers on the Pd cubic or octahedral seeds. We use the epitaxial growth of PtRuFeCoNi atomic layers as an example to introduce the standard protocol of epitaxial growth in this work. A total of 0.5 ml of OAm solution containing Pd seeds is mixed with 3 ml of ODE in a 23-ml vial. Then, the mixture is preheated at 110°C for 20 min under magnetic stirring, followed by an increase to 200°C for 20 min. Subsequently, a 14-ml precursor solution containing $H_2PtCl_6{\cdot}6H_2O$, $RuCl_3{\cdot}xH_2O$, $Ni(C_5H_7O_2)_2$, $Co(C_5H_7O_2)_2$, and $Fe(C_5H_7O_2)_3$, each with an equal molar concentration of 0.042 μmol ml^{-1} in OAm, is prepared. Using a syringe pump, this precursor solution is then added dropwise to the reaction mixture containing Pd seeds at a constant injection rate of 0.8 ml $hour^{-1}$. After the precursor solution has been added, the reaction mixture is kept at 200°C for another 1 hour. After that, the Pd@PtRuFeCoNi core-shell nanocrystals are collected through centrifugation and washing with *n*-hexane three times and redispersed in DI water. To synthesize other quinary atomic layers on the Pd seeds, we follow a similar protocol as described above, except for the used precursors. In these experiments, metal precursors including $Pd(C_5H_7O_2)_2$, $H_2IrCl_6{\cdot}xH_2O$, and $RhCl_3{\cdot}xH_2O$ are selected for Pd, Ir, and Rh elements, respectively.

Electrocatalytic measurements for HER

In this work, we conduct the HER electrochemical measurements using a Nova electrochemical workstation (Metrohm Autolab PGSTAT204). A glassy carbon cell is used as the working electrode for rotating electrodes, with Pt as the counter electrode and Ag/AgCl as the reference electrode. To prepare the catalyst ink, 1 mg of Pd@IGM-PGM-HEA core-shell nanocubes on the carbon support (Vulcan XC-72R) with 20 wt % is dispersed in a mixture of 0.5 ml of isopropanol, 0.5 ml of DI water, and 0.02 ml of a 5% Nafion solution. To facilitate comparative analysis, an equivalent loading of commercial 20 wt % Pt nanoparticles on Vulcan XC-72 (Pt/C; Premetek, USA), featuring an average size of approximately ~3 nm and supported on carbon black, is also incorporated in 1 mg of catalysts. We have verified the 20 wt % metal loading through ICP-OES analysis in preparation for subsequent electrocatalytic measurements. The resulting mixture is subjected to an ice-cooled ultrasonic bath for homogeneous catalyst dispersion. Subsequently, 3.6 μl of the prepared catalyst solution (0.01 mg/cm^2 geo) is dispersed onto a glassy carbon electrode with a geometric area of 0.07 cm^2. The electrode is then dried under an atmosphere saturated with isopropanol before the electrochemical measurements. Electrochemical characterizations for HER investigations are performed in 0.5 M H_2SO_4 electrolytes saturated with argon gas. First, 150 cycles of cyclic voltammetry (CV) are performed using the high scan rate of 500 mV s^{-1} within a potential range of 0.05 to 1.05 V versus RHE to establish a stable catalyst layer. Following the activation step, an LSV measurement is conducted by scanning the potential from 0.1 to −0.25 V (1600 rpm, 1 mV s^{-1}) to evaluate the HER performance. The durability evaluation of catalysts is conducted through extensive cycling (15,000 cycles, 100 mV s^{-1}) within an identical potential range as the LSV procedure described above.

Electrocatalytic measurements for HOR

The catalyst preparation and the loading quantity of catalysts onto the carbon support (Vulcan XC-72R) remain consistent with the previously outlined procedure for HER electrocatalytic measurements. We use Hg/HgO electrode as the reference electrode in a 0.1 M KOH electrolyte. The electrolyte is first purged with N_2 gas for 30 min before electrocatalytic measurements to remove impurities. Subsequently, the electrode surface is cleaned by 50 cycles of CV test in the potential region of 0.05 to 1.05 V at a high scan speed (500 mV s^{-1}). The measurements are performed in N_2 and H_2 atmospheres, respectively. The CV and LSV curves are recorded separately from −0.05 to 0.7 V (1600 rpm) with a scan rate of 10 and 5 mV s^{-1}. The accelerated durability tests are carried out in the same region at a scan rate of 500 mV s^{-1} for 3000 cycles in the H_2-saturated electrolyte.

Measurements and estimation of ECSA

In this work, the ECSAs of catalysts are calculated and measured experimentally by using the following two methods.

Method 1: Calculation the ECSA of catalysts

We estimate the ECSA of the two types of catalysts including the commercial Pt/C (Premetek, USA) and Pd@IGM-PGM-HEA core-shell nanocubes. For the commercial Pt/C catalysts, we assume that the shape of 3-nm Pt nanoparticles in commercial Pt/C is a perfect sphere. For the Pd@IGM-PGM-HEA core-shell nanocubes, we assume that the 18.4-nm Pd@IGM-PGM-HEA core-shell nanocubes maintain their ideal cubic shape after the epitaxial growth (Fig. 3). Therefore, the whole ECSAs of these two catalysts are calculated using the following equations (*53*):

$$Surface\ area\ of\ Pt\ \ nanoparticle\ in\ \ Pt/C = 4\pi r^2 \tag{2}$$

where r is the radius of the nanoparticle;

$$Surface\ area\ of\ Pd@IGM-PGM-HEA\ core-shell\ nanocube = 6l^2 \tag{3}$$

where l is the edge length of nanocube;

$$ECSA = \frac{weight\ of\ metal\ loading\ on\ electrode}{weight\ of\ each\ nanoparticle\ or\ nanocube} \times surface\ area\ of\ nanoparticle\ or\ nanocube \tag{4}$$

Method 2: Double-layer capacitance

The double-layer capacitance (C_{dl}) is measured by CV with scan rates ranging from 25 to 500 mV s^{-1} in a 0.5 M N_2-saturated H_2SO_4 electrolyte. The potential window of 0.1 V, centered at the open circuit potential, is selected for the non-faradic region. In the evaluation of C_{dl}, the currents of cathodic and anodic are divided by the respective scan rates to obtain the resulting slope (*54*). The ECSA is subsequently calculated using the C_{dl} and a selected value of specific capacitance (C_s) according to the equation

$$ECSA = \frac{Cdl}{Cs} \tag{5}$$

X-ray absorption spectroscopy

We obtain the XAS spectra using a customized three-electrode system with a fluorescence mode. The system consists of a Teflon

container equipped with a window covered by Kapton tape. The experimental conditions for XAS measurements are identical to those described above for electrocatalytic acidic HER measurements. During the XAS measurements, x-rays pass through the Kapton tape, and total fluorescence-yield mode detection is used. These measurements are performed using Lytle detectors at the BL-17C and 01C1 beamlines for HER and the TPS 44A beamline for HOR at the National Synchrotron Radiation Research Center (Hsinchu, Taiwan). The obtained data are analyzed using Athena software to examine XANES of the K-edges of Ru, Fe, Co, and Ni, and the L_3-edge of Pt. Furthermore, FT-EXAFS analysis is conducted by using Fourier transform on k^2-weighted oscillations EXAFS to assess the local coordination environments of the target elements.

TEM, HAADF-STEM, and EDS

The size and morphological features of the obtained nanocrystals are characterized by TEM (JEM-2100 Electron Microscope), operating at an acceleration voltage of 200 kV. For the preparation of TEM samples, a droplet of 0.5 μl aqueous dilute solutions containing the obtained nanocrystals is drop-casted on the carbon-coated 200-mesh copper grids through a pipette and then dried in an oven at 60°C for 10 min. In particular, the Pd@IGM-PGM-HEA core-shell nanocrystals consisting of Fe, Co, and Ni magnetic elements are demagnetized through a demagnetizer (Holding Standard Type Demagnetizer HD-200) before the TEM imaging. After the TEM analysis, the size of the nanocrystals is measured using the ImageJ software.

The surface atomic arrangements of the Pd@IGM-PGM-HEA core-shell nanocrystals are characterized by atomic-resolution HAADF-STEM (Spherical-Aberration Corrected Field Emission TEM, JEOL JEM-ARM200FTH). The aqueous suspension of the Pd@IGM-PGM-HEA core-shell nanocrystals is dropped on the carbon-coated 200-mesh copper grids and then dried in an oven at 60°C for 10 min after a cleaning procedure involving successive centrifugation and washing cycles with DI water and absolute ethanol for five times, respectively. All HAADF-STEM images with a dark-field lattice image resolution of ~0.08 nm are obtained using an acceleration voltage of 200 kV. Through the HAADF-STEM imaging mode, larger atomic numbers of elements can scatter electrons more substantially, leading to a Z-contrast image. This allows us to resolve the elemental distinction between the Pd core and the IGM-PGM-HEA shell with different atomic numbers.

For the elemental analysis and the spatial distribution of the constituent elements, the Pd@IGM-PGM-HEA core-shell nanocrystals are mapped using EDS on a cold-field emission JEOL JEM-ARM200FTH aberration-corrected STEM. The Oxford Aztec EDS system is used for the EDS analysis, which has an 80-mm^2 silicon drift detector (detector model: Ultim Max TEM; liquid nitrogen–free operation) with ≦129-eV resolution. The EDS elemental mapping images with a spectrum range spanning from 0 to 40 keV are acquired under the following experimental conditions: The microscope is operated at 200 kV, with an STEM spot size of 5C and a camera length of 6 cm. In general, the elemental signals of the Pd@IGM-PGM-HEA core-shell nanocrystals are collected for around 5 min at a count rate of approximately 150 counts/s. Please note that during the process of revising the manuscript, the EDS analysis system had an upgrade, and the count rate is thus elevated to around 500 counts/s for the Pd@PtPdFeCoNi, Pd@PtRuFeCoNi, Pd@IrPdFeCoNi, Pd@IrRuFeCoNi, and Pd@PdRuFeCoNi atomic layers (Fig. 3 and figs. S5, S6, S8, and S14). The subsequent data processing of the acquired EDS mappings is through the Aztec software. The contrast, brightness, and false coloring are adjusted in the Aztec software for ensuring optimal viewing.

To confirm the metastable phase of IGM-PGM-HEA atomic layers on the Pd cubic seeds as well as to reveal their equilibration pathways, in situ heating experiments are carried out through a heating holder (Protochips Fusion 350) and the corresponding thermal E-chips (Protochips) capable of uniformly heating samples up to 1200°C with any heating rate up to 1000°C ms^{-1} inside the JEOL JEM-ARM200FTH aberration-corrected STEM operated at 200 kV. The aqueous suspension of the Pd@IGM-PGM-HEA core-shell nanocrystals is dropped on the thermal E-chips and then allowed to fully air-dry before further use. Subsequently, the thermal E-chip is incorporated with the in situ heating holder. The temperature elevation follows a heating rate of 60°C min^{-1} under vacuum and is maintained at room temperature, 100°, 200°, 300°, 400°, and 500°C, each for a duration of 10 min. We obtain HAADF-STEM images of the Pd@IGM-PGM-HEA core-shell nanocrystals at these temperatures during heating up to 500°C under vacuum.

Inductively coupled plasma optical emission spectroscopy

The elemental compositions of our catalysts (Pt, Ir, Pd, Rh, Ru, Fe, Co, and Ni) are measured by ICP-OES (Thermo Fisher Scientific iCAP 7200). The samples are dissolved in aqua regia, followed by recovery within a 1% HNO_3 solution before the measurement process.

Characterizations

We analyze the element contents using an ICP-OES (Thermo Fisher Scientific iCAP 7200). The morphologies of samples are characterized by TEM and HAADF-STEM. EDS mapping is performed on a spherical-aberration corrected field emission TEM (JEOL, JEM-ARM200FTH) operated at 200 kV. We analyze the crystal structures using XRD on an x-ray diffractometer (Bruker, D8). We study surface chemistry through XPS with a high-resolution s-ray photoelectron spectrometer (ULVAC-PHI, PHI QuanteraII).

DFT computations

All computations were performed using DFT with the revised Perdew-Burke-Ernzerhof functional (*55*), as implemented in the Vienna Ab nitio Simulation Package (*56*). The valence electron wave functions were expanded in plane-wave basis sets with a 380-eV cutoff, and the projector augmented wave method was used to describe the core-electron interactions (*57*). A 7 × 7 × 1 k-point grid was sampled using the Monkhorst-Pack scheme for each computation. All structures were fully optimized until the residual forces on the constituent atoms became less than 0.02 eV/Å while keeping the lattice parameter fixed, which was taken from the experiment. For further computational details, please refer to the "Computational details" section in the Supplementary Materials.

Supplementary Materials

This PDF file includes:

Figs. S1 to S29
Tables S1 to S7
References

REFERENCES AND NOTES

1. J.-W. The, S.-K. Chen, S.-J. Lin, J.-Y. Gan, T.-S. Chin, T.-T. Shun, C.-H. Tsau, S.-Y. Chang, Nanostructured high-entropy alloys with multiple principal elements: Novel alloy design concepts and outcomes. *Adv. Eng. Mater.* **6**, 299–303 (2004).

2. Y. Yao, Q. Dong, A. Brozena, J. Luo, J. Miao, M. Chi, C. Wang, I. Keverkidis, Z. J. Ren, J. Greeley, G. Wang, A. Anapolsky, L. Hu, High-entropy nanoparticles: Synthesis-structure-property relationships and data-driven discovery. *Science* **376**, eabn3103 (2022).
3. Y. Yao, Z. Huang, P. Xie, S. D. Lacey, R. J. Jacob, H. Xie, F. Chen, A. Nie, T. Pu, M. Rehwoldt, D. Yu, M. R. Zachariah, C. Wang, R. Shahbazian-Yassar, J. Li, L. Hu, Carbothermal shock synthesis of high-entropy-alloy nanoparticles. *Science* **359**, 1489–1494 (2018).
4. P. Xie, Y. Yao, Z. Huang, Z. Liu, J. Zhang, T. Li, G. Wang, R. Shahbazian-Yassar, L. Hu, C. Wang, Highly efficient decomposition of ammonia using high-entropy alloy catalysts. *Nat. Commun.* **10**, 4011 (2019).
5. X. Wang, Q. Dong, H. Qiao, Z. Huang, M. T. Saray, G. Zhong, Z. Lin, M. Cui, A. Brozena, M. Hong, Q. Xia, J. Gao, G. Chen, R. Shahbazian-Yassar, D. Wang, L. Hu, Continuous synthesis of hollow high-entropy nanoparticles for energy and catalysis applications. *Adv. Mater.* **32**, e2002853 (2020).
6. H. Li, Y. Han, H. Zhao, W. Qi, D. Zhang, Y. Yu, W. Cai, S. Li, J. Lai, B. Huang, L. Wang, Fast site-to-site electron transfer of high-entropy alloy nanocatalyst driving redox electrocatalysis. *Nat. Commun.* **11**, 5437 (2020).
7. S. Gao, S. Hao, Z. Huang, Y. Yuan, S. Han, L. Lei, X. Zhang, R. Shahbazian-Yassar, J. Lu, Synthesis of high-entropy alloy nanoparticles on supports by the fast moving bed pyrolysis. *Nat. Commun.* **11**, 2016 (2020).
8. D. Wu, K. Kusada, T. Yamamoto, T. Toriyama, S. Matsumura, S. Kawaguchi, Y. Kubota, H. Kitagawa, Platinum-group-metal high-entropy-alloy nanoparticles. *J. Am. Chem. Soc.* **142**, 13833–13838 (2020).
9. B. Song, Y. Yang, M. Rabbani, T. T. Yang, K. He, X. Hu, Y. Yuan, P. Ghildiyal, V. P. Dravid, M. R. Zachariah, W. A. Saidi, Y. Liu, In situ oxidation studies of high-entropy alloy nanoparticles. *ACS Nano* **14**, 15131–15143 (2020).
10. G. Feng, F. Ning, J. Song, H. Shang, K. Zhang, Z. Ding, P. Gao, W. Chu, D. Xia, Sub-2 nm ultrasmall high-entropy alloy nanoparticles for extremely superior electrocatalytic hydrogen evolution. *J. Am. Chem. Soc.* **143**, 17117–17127 (2021).
11. C. Zhan, Y. Xu, L. Bu, H. Zhu, Y. Feng, T. Yang, Y. Zhang, Z. Yang, B. Huang, Q. Shao, X. Huang, Subnanometer high-entropy alloy nanowires enable remarkable hydrogen oxidation catalysis. *Nat. Commun.* **12**, 6261 (2021).
12. T. Li, Y. Yao, Z. Huang, P. Xie, Z. Liu, M. Yang, J. Gao, K. Zeng, A. H. Brozena, G. Pastel, M. Jiao, Q. Dong, J. Dai, S. Li, H. Zong, M. Chi, J. Luo, Y. Mo, G. Wang, C. Wang, R. Shahbazian-Yassar, L. Hu, Denary oxide nanoparticles as highly stable catalysts for methane combustion. *Nat. Catal.* **4**, 62–70 (2021).
13. C.-L. Yang, L.-N. Wang, P. Yin, J. Liu, M.-X. Chen, Q.-Q. Yan, Z.-S. Wang, S.-L. Xu, S.-Q. Chu, C. Cui, H. Ju, J. Zhu, Y. Lin, J. Shui, H.-W. Liang, Sulfur-anchoring synthesis of platinum intermetallic nanoparticle catalysts for fuel cells. *Science* **374**, 459–464 (2021).
14. K. Mori, N. Hashimoto, N. Kamiuchi, H. Yoshida, H. Kobayashi, H. Yamashita, Hydrogen spillover-driven synthesis of high-entropy alloy nanoparticles as a robust catalyst for CO_2 hydrogenation. *Nat. Commun.* **12**, 3884 (2021).
15. E. Shen, X. Song, Q. Chen, M. Zheng, J. Bian, H. Liu, Spontaneously forming oxide layer of high entropy alloy nanoparticles deposited on porous carbons for supercapacitors. *ChemElectroChem* **8**, 260–269 (2021).
16. M. Cui, C. Yang, S. Huang, M. Yang, S. Overa, Q. Dong, Y. Yao, A. H. Brozena, D. A. Cullen, M. Chi, T. F. Blum, D. Morris, Z. Finfrock, X. Wang, P. Zhang, V. G. Goncharov, X. Guo, J. Luo, Y. Mo, F. Jiao, L. Hu, Multi-principal elemental intermetallic nanoparticles synthesized via a disorder-to-order transition. *Sci. Adv.* **8**, eabm4322 (2022).
17. H. Minamihara, K. Kusada, D. Wu, T. Yamamoto, T. Toriyama, S. Matsumura, L. S. R. Kumara, K. Ohara, O. Sakata, S. Kawaguchi, Y. Kubota, H. Kitagawa, Continuous-flow reactor synthesis for homogeneous 1 nm-sized extremely small high-entropy alloy nanoparticles. *J. Am. Chem. Soc.* **144**, 11525–11529 (2022).
18. S. L. Bueno, A. Leonardi, N. Kar, K. Chatterjee, X. Zhan, C. Chen, Z. Wang, M. Engel, V. Fung, S. E. Skrabalak, Quinary, senary, and septenary high entropy alloy nanoparticle catalysts from core@shell nanoparticles and the significance of intraparticle heterogeneity. *ACS Nano* **16**, 18873–18885 (2022).
19. G. Gao, J. Liang, Z. Guo, K. Yang, G. Wang, H. Wang, X. Wan, Z. Li, Y. Bai, Y. Zhang, J. Liu, Y. Feng, Z. Zheng, C. Lu, G. He, Z. Xiong, Z. Liu, S. Chen, Y. Guo, M. Zeng, J. Lin, L. Fu, Liquid metal for high-entropy alloy nanoparticles synthesis. *Nature* **619**, 73–77 (2023).
20. Y.-H. Liu, C.-J. Hsieh, L.-C. Hsu, K.-H. Lin, Y.-C. Hsiao, C.-C. Chi, J.-T. Lin, C.-W. Chang, S.-C. Lin, C.-W. Wu, J.-Q. Gao, C.-W. Pao, Y.-M. Chang, M.-Y. Lu, S. Zhou, T.-H. Yang, Toward controllable and predictable synthesis of high-entropy alloy nanocrystals. *Sci. Adv.* **9**, eadf9931 (2023).
21. G. Feng, F. Ning, Y. Pan, T. Chen, J. Song, Y. Wang, R. Zou, D. Su, D. Xia, Engineering structurally ordered high-entropy intermetallic nanoparticles with high-activity facets for oxygen reduction in practical fuel cells. *J. Am. Chem. Soc.* **145**, 11140–11150 (2023).
22. B. Pang, X. Liu, T. Liu, T. Chen, X. Shen, W. Zhang, S. Wang, T. Liu, D. Liu, T. Ding, Z. Liao, Y. Li, C. Liang, T. Yao, Laser-assisted high-performance PtRu alloy for pH-universal hydrogen evolution. *Energy Environ. Sci.* **15**, 102–108 (2022).
23. D. Zhang, H. Zhao, B. Huang, B. Li, H. Li, Y. Han, Z. Wang, X. Wu, Y. Pan, Y. Sun, X. Sun, J. Lai, L. Wang, Advanced ultrathin RuPdM (M = Ni, Co, Fe) nanosheets electrocatalyst boosts hydrogen evolution. *ACS Cent. Sci.* **5**, 1991–1997 (2019).
24. X. Chen, S. Zhang, C. Li, Z. Liu, X. Sun, S. Cheng, D. N. Zakharov, S. Hwang, Y. Zhu, J. Fang, G. Wang, G. Zhou, Composition-dependent ordering transformations in Pt–Fe nanoalloys. *Proc. Natl. Acad. Sci. U.S.A.* **119**, e2117899119 (2022).
25. Q. Gao, B. Yao, H. S. Pillai, W. Zang, X. Han, Y. Liu, S. W. Yu, Z. Yan, B. Min, S. Zhang, H. Zhou, L. Ma, H. Xin, Q. He, H. Zhu, Synthesis of core/shell nanocrystals with ordered intermetallic single-atom alloy layers for nitrate electroreduction to ammonia. *Nat. Synth.* **2**, 624–634 (2023).
26. H. Chen, G. Wang, T. Gao, Y. Chen, H. Liao, X. Guo, H. Li, R. Liu, M. Dou, S. Nan, Q. He, Effect of atomic ordering transformation of PtNi nanoparticles on alkaline hydrogen evolution: Unexpected superior activity of the disordered phase. *J. Phys. Chem. C* **124**, 5036–5045 (2020).
27. H. Li, S. Xu, M. Wang, Z. Chen, F. Ji, K. Cheng, Z. Gao, Z. Ding, W. Yang, Computational design of (100) alloy surfaces for the hydrogen evolution reaction. *J. Mater. Chem. A* **8**, 17987–17997 (2020).
28. T. Wu, M. L. Stone, M. J. Shearer, M. J. Stolt, L. A. Guzei, R. J. Hamers, R. Lu, K. Deng, S. Jin, J. R. Schmidt, Crystallographic facet dependence of the hydrogen evolution reaction on cops: Theory and experiments. *ACS Catal.* **8**, 1143–1152 (2018).
29. X. Guo, T. Wu, S. Zhao, Y. Fang, S. Xu, M. Xie, F. Huang, Re modulation of metallic ultrathin 2m-ws_2 for highly efficient hydrogen evolution in both acidic and alkaline media. *ACS Appl. Energy Mater.* **5**, 7674–7680 (2022).
30. Y. Niu, X. Qian, C. Xu, H. Liu, W. Wu, L. Hou, Cu-Ni-$CoSe_x$ quaternary porous nanocubes as enhanced Pt-free electrocatalysts for highly efficient dye-sensitized solar cells and hydrogen evolution in alkaline medium. *Chem. Eng. J.* **357**, 11–20 (2019).
31. Z. Ye, Y. Qie, Z. Fan, Y. Liu, Z. Shi, H. Yang, Soft magnetic Fe_5C_2–Fe_3C@C as an electrocatalyst for the hydrogen evolution reaction. *Dalton Trans.* **48**, 4636–4642 (2019).
32. C. Wang, Z. Huang, Y. Ding, M. Xie, M. Chi, Y. Xia, Facet-controlled synthesis of platinum-group-metal quaternary alloys: The case of nanocubes and {100} facets. *J. Am. Chem. Soc.* **145**, 2553–2560 (2023).
33. J. Park, L. Zhang, S. Choi, L. T. Roling, N. Lu, J. A. Herron, S. Xie, J. Wang, M. J. Kim, M. Mavrikakis, Y. Xia, Atomic layer-by-layer deposition of platinum on palladium octahedra for enhanced catalysts toward the oxygen reduction reaction. *ACS Nano* **9**, 2635–2647 (2015).
34. Y. Liao, *Practical electron microscopy and database* (www.globalsino.com/EM/) (2006).
35. D. C. Bock, C. J. Pelliccione, W. Zhang, J. Timoshenko, K. W. Knehr, A. C. West, F. Wang, Y. Li, A. I. Frenkel, E. S. Takeuchi, K. J. Takeuchi, A. C. Marschilok, Size dependent behavior of Fe_3O_4 crystals during electrochemical (de) lithiation: An in situ x-ray diffraction, ex situ x-ray absorption spectroscopy, transmission electron microscopy and theoretical investigation. *Phys. Chem. Chem. Phys.* **19**, 20867–20880 (2017).
36. Q. Li, H. Zhu, X. Chen, H. Liu, Y. Ren, Y. Chen, K. Ohara, L. Zhou, J. Chen, J. Deng, J. Miao, K. Lin, X. Kuang, X. Xing, Local structure insight into hydrogen evolution reaction with bimetal nanocatalysts. *J. Am. Chem. Soc.* **144**, 20298–20305 (2022).
37. H. Sepehri-Amin, H. Iwama, T. Ohkubo, T. Shima, K. Hono, Microstructure and in-plane component of $L1_0$-FePt films deposited on MgO and $MgAl_2O_4$ substrates. *Scr. Mater.* **130**, 247–251 (2017).
38. F. Li, Y. Zong, Y. Ma, M. Wang, W. Shang, P. Tao, C. Song, T. Deng, H. Zhu, J. Wu, Atomistic imaging of competition between surface diffusion and phase transition during the intermetallic formation of faceted particles. *ACS Nano* **15**, 5284–5293 (2021).
39. Q. Yun, Y. Ge, Z. Shi, J. Liu, X. Wang, A. Zhang, B. Huang, Y. Yao, Q. Luo, L. Zhai, J. Ge, Y. Peng, C. Gong, M. Zhao, Y. Qin, C. Ma, G. Wang, Q. Wa, X. Zhou, Z. Li, S. Li, W. Zhai, H. Yang, Y. Ren, Y. Wang, L. Li, X. Ruan, Y. Wu, B. Chen, Q. Lu, Z. Lai, Q. He, X. Huang, Y. Chen, H. Zhang, Recent progress on phase engineering of nanomaterials. *Chem. Rev.* **123**, 13489–13692 (2023).
40. D. R. Dekel, Review of cell performance in anion exchange membrane fuel cells. *J. Power Sources* **375**, 158–169 (2018).
41. M. Carmo, D. L. Fritz, J. Mergel, D. Stolten, A comprehensive review on PEM water electrolysis. *Int. J. Hydrogen Energy* **38**, 4901–4934 (2013).
42. H. Sun, X. Xu, H. Kim, W. Jung, W. Zhou, Z. Shao, Electrochemical water splitting: Bridging the gaps between fundamental research and industrial applications. *Energy Environ. Mater.* **6**, e12441 (2023).
43. Y. Xia, X. Xia, H. C. Peng, Shape-controlled synthesis of colloidal metal nanocrystals: Thermodynamic versus kinetic products. *J. Am. Chem. Soc.* **137**, 7947–7966 (2015).
44. L. Han, P. Ou, W. Liu, X. Wang, H. T. Wang, R. Zhang, C. W. Pao, X. Liu, W. F. Pong, J. Song, Z. Zhuang, M. V. Mirkin, J. Luo, H. L. Xin, Design of Ru-Ni diatomic sites for efficient alkaline hydrogen oxidation. *Sci. Adv.* **8**, eabm3779 (2022).
45. L. Jiao, E. Liu, S. Hwang, S. Mukerjee, Q. Jia, Compressive strain reduces the hydrogen evolution and oxidation reaction activity of platinum in alkaline solution. *ACS Catal.* **11**, 8165–8173 (2021).
46. Q. He, D. Tian, H. Jiang, D. Cao, S. Wei, D. Liu, P. Song, Y. Lin, L. Song, Achieving efficient alkaline hydrogen evolution reaction over a Ni_5P_4 catalyst incorporating single-atomic Ru sites. *Adv. Mater.* **32**, e1906972 (2020).
47. L. Zhang, R. Si, H. Liu, N. Chen, Q. Wang, K. Adair, Z. Wang, J. Chen, Z. Song, J. Li, M. N. Banis, R. Li, T. K. Sham, M. Gu, L. M. Liu, G. A. Botton, X. Sun, Atomic layer deposited

Pt-Ru dual-metal dimers and identifying their active sites for hydrogen evolution reaction. *Nat. Commun.* **10**, 4936 (2019).
48. T. Tang, X. Liu, X. Luo, Z. Xue, H. R. Pan, J. Fu, Z. C. Yao, Z. Jiang, Z. H. Lyu, L. Zheng, D. Su, J. N. Zhang, L. Zhang, J. S. Hu, Unconventional bilateral compressive strained Ni–Ir interface synergistically accelerates alkaline hydrogen oxidation. *J. Am. Chem. Soc.* **145**, 13805–13815 (2023).
49. Z. C. Yao, T. Tang, Z. Jiang, L. Wang, J. S. Hu, L. J. Wan, Electrocatalytic hydrogen oxidation in alkaline media: From mechanistic insights to catalyst design. *ACS Nano* **16**, 5153–5183 (2022).
50. X. Wang, S. I. Choi, L. T. Roling, M. Luo, C. Ma, L. Zhang, M. Chi, J. Liu, Z. Xie, J. A. Herron, M. Mavrikakis, Y. Xia, Palladium–platinum core-shell icosahedra with substantially enhanced activity and durability towards oxygen reduction. *Nat. Commun.* **6**, 7594 (2015).
51. J. Li, J. Hu, M. Zhang, W. Gou, S. Zhang, Z. Chen, Y. Qu, Y. Ma, A fundamental viewpoint on the hydrogen spillover phenomenon of electrocatalytic hydrogen evolution. *Nat. Commun.* **12**, 3502 (2021).
52. B. Lim, M. Jiang, J. Tao, P. H. Camargo, Y. Zhu, Y. Xia, Shape-controlled synthesis of Pd nanocrystals in aqueous solutions. *Adv. Funct. Mater.* **19**, 189–200 (2009).
53. W. Zhu, S. Kattel, F. Jiao, J. G. Chen, Shape-controlled CO_2 electrochemical reduction on nanosized pd hydride cubes and octahedra. *Adv. Energy Mater.* **9**, 1802840 (2019).
54. D. M. Morales, M. Risch, Seven steps to reliable cyclic voltammetry measurements for the determination of double layer capacitance. *J. Phys. Energy* **3**, 034013 (2021).
55. B. Hammer, L. B. Hansen, J. K. Nørskov, Improved adsorption energetics within density-functional theory using revised Perdew-Burke-Ernzerhof functionals. *Phys. Rev. B* **59**, 7413–7421 (1999).
56. G. Kresse, J. Furthmüller, Efficient iterative schemes for ab initio total-energy calculations using a plane-wave basis set. *Phys. Rev. B* **54**, 11169–11186 (1996).
57. G. Kresse, D. Joubert, From ultrasoft pseudopotentials to the projector augmented-wave method. *Phys. Rev. B* **59**, 1758–1775 (1999).
58. X. Xia, S. Xie, M. Liu, H. C. Peng, N. Liu, J. Wang, M. J. Kim, Y. Xia, On the role of surface diffusion in determining the shape or morphology of noble-metal nanocrystals. *Proc. Natl. Acad. Sci. U.S.A.* **110**, 6669–6673 (2013).
59. K. D. Gilroy, X. Yang, S. Xie, M. Zhao, D. Qin, Y. Xia, Shape-controlled synthesis of colloidal metal nanocrystals by replicating the surface atomic structure on the seed. *Adv. Mater.* **30**, e1706312 (2018).
60. F. Xing, J. Ma, K. I. Shimizu, S. Furukawa, High-entropy intermetallics on ceria as efficient catalysts for the oxidative dehydrogenation of propane using CO_2. *Nat. Commun.* **13**, 5065 (2022).
61. H. Sun, C. W. Tung, Y. Qiu, W. Zhang, Q. Wang, Z. Li, J. Tang, H. C. Chen, C. Wang, H. M. Chen, Atomic metal–support interaction enables reconstruction-free dual-site electrocatalyst. *J. Am. Chem. Soc.* **144**, 1174–1186 (2022).
62. Q. Yin, J. Luo, J. Zhang, L. Zheng, G. Cui, J. Han, D. O'Hare, High-performance, long lifetime chloride ion battery using a NiFe-Cl layered double hydroxide cathode. *J. Mater. Chem. A* **8**, 12548–12555 (2020).
63. H. Gunda, S. K. Das, K. Jasuja, Simple, green, and high-yield production of boron-based nanostructures with diverse morphologies by dissolution and recrystallization of layered magnesium diboride crystals in water. *ChemPhysChem* **19**, 880–891 (2018).
64. X. Wang, R. R. D. Vecchis, C. Li, H. Zhang, X. Hu, S. Sridar, Y. Wang, W. Chen, W. Xiong, Design metastability in high-entropy alloys by tailoring unstable fault energies. *Sci. Adv.* **8**, eabo7333 (2022).
65. C. C. McCrory, S. Jung, J. C. Peters, T. F. Jaramillo, Benchmarking heterogeneous electrocatalysts for the oxygen evolution reaction. *J. Am. Chem. Soc.* **135**, 16977–16987 (2013).
66. J. K. Norskov, T. Bligaard, A. Logadottir, J. R. Kitchin, J. G. Chen, S. Pandelov, U. Stimming, Trends in the exchange current for hydrogen evolution. *J. Electrochem. Soc.* **152**, J23 (2005).
67. J. Greeley, T. F. Jaramillo, J. Bonde, I. Chorkendorff, J. K. Norskov, Computational high-throughput screening of electrocatalytic materials for hydrogen evolution. *Nat. Mater.* **5**, 909–913 (2006).
68. Y. Li, Y. Huang, Low-temperature, seed-mediated synthesis of monodispersed hyperbranched PtRu nanoparticles and their electrocatalytic activity in methanol oxidation. *J. Mater. Chem.* **22**, 12461–12464 (2012).
69. J. Park, T. Kwon, J. Kim, H. Jin, H. Y. Kim, B. Kim, S. H. Joo, K. Lee, Hollow nanoparticles as emerging electrocatalysts for renewable energy conversion reactions. *Chem. Soc. Rev.* **47**, 8173–8202 (2018).
70. H. C. Peng, S. Xie, J. Park, X. Xia, Y. Xia, Quantitative analysis of the coverage density of Br^- ions on Pd {100} facets and its role in controlling the shape of Pd nanocrystals. *J. Am. Chem. Soc.* **135**, 3780–3783 (2013).
71. S. Xie, S. Choi, N. Lu, L. T. Roling, J. A. Herron, L. Zhang, J. Wang, M. J. Kim, Z. Xie, M. Mavrikakis, Y. Xia, Atomic layer-by-layer deposition of Pt on Pd nanocubes for catalysts with enhanced activity and durability toward oxygen reduction. *Nano Lett.* **14**, 3570–3576 (2014).
72. J. F. Moulder, W. F. Stickle, P. E. Sobol, K. D. Bomben, *Handbook of X-ray Photoelectron Spectroscopy* (Perkin-Elmer, 1992).
73. J. A. Ramos-Guivar, E. A. Sanches, F. Bruns, E. Sadrollahi, M. A. Morales, E. O. Lopez, F. J. Litterst, Vacancy ordered γ-Fe2O3 nanoparticles functionalized with nanohydroxyapatite: XRD, FTIR, TEM, XPS and Mössbauer studies. *Appl. Surf. Sci.* **389**, 721–734 (2016).
74. C. Pirlot, J. Delhalle, J. J. Pireaux, Z. Mekhalif, Surface modification of polycrystalline iron surfaces by n-dodecanethiol: An XPS investigation. *Surf. Coat. Technol.* **138**, 166–172 (2001).
75. F. A. Stevie, C. L. Donley, Introduction to x-ray photoelectron spectroscopy. *J. Vac. Sci. Technol. A* **38**, 063204 (2020).

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